

Line-by-Line Mathematical Guide for HAETAE_Rejection.ec

Generated HAETAE EasyCrypt proof guide

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File Role

Rejection predicates and rejection-sampling loss accounting.

How To Read This Script

Read this file as the link between accepted HAETAE samples, transcript validity, and the rejection loss terms used in the final theorem.

Line-by-Line Mathematical Reading

Each row preserves one EasyCrypt source line and gives the intended mathematical or proof-script reading. Blank rows are kept because they mark proof structure in long EasyCrypt developments.

Line	EasyCrypt source	Mathematical reading
1	<code>require import AllCore Distr IntDiv List Real StdOrder RealSeries.</code>	Imports AllCore Distr IntDiv List Real StdOrder RealSeries. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
2	<code>require import HAETAE_Params HAETAE_Algebra HAETAE_Distributions.</code>	Imports HAETAE_Params HAETAE_Algebra HAETAE_Distributions. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
3	<code>require import HAETAE_Events HAETAE_Reductions.</code>	Imports HAETAE_Events HAETAE_Reductions. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
4	<code>require import HAETAE_Transcript.</code>	Imports HAETAE_Transcript. Mathematically, this exposes earlier carrier sets, operations, games, and lemmas that the current file may cite.
5	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
6	<code>theory HAETAE_Rejection.</code>	Begins the namespace HAETAE_Rejection, grouping the declarations as one checked theory.
7	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
8	<code>import HAETAE_Params.</code>	Opens HAETAE_Params so its names can be used without qualification in the following proof.
9	<code>import HAETAE_Algebra.</code>	Opens HAETAE_Algebra so its names can be used without qualification in the following proof.
10	<code>import HAETAE_Distributions.</code>	Opens HAETAE_Distributions so its names can be used without qualification in the following proof.
11	<code>import HAETAE_Events.</code>	Opens HAETAE_Events so its names can be used without qualification in the following proof.
12	<code>import HAETAE_Reductions.</code>	Opens HAETAE_Reductions so its names can be used without qualification in the following proof.
13	<code>import HAETAE_Transcript.</code>	Opens HAETAE_Transcript so its names can be used without qualification in the following proof.

Line	EasyCrypt source	Mathematical reading
14	<code>import RealOrder.</code>	Opens RealOrder so its names can be used without qualification in the following proof.
15	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
16	<code>type signing_transcript_record = message * context * signature * transcript.</code>	Declares carrier set <code>signing_transcript_record</code> . EasyCrypt will type-check all later functions and games over this mathematical domain.
17	<code>type signing_attempt_sample = polyvecl * polyveck * polyveck.</code>	Declares carrier set <code>signing_attempt_sample</code> . EasyCrypt will type-check all later functions and games over this mathematical domain.
18	<code>type signing_attempt_state =</code>	Declares carrier set <code>signing_attempt_state</code> . EasyCrypt will type-check all later functions and games over this mathematical domain.
19	<code>random_coins * signing_attempt_sample * polyveck * poly * challenge *</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
20	<code>polyvecl * polyveck * hint_t * signature.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
21	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
22	<code>op signing_attempt_state_coins</code>	Defines operation <code>signing_attempt_state_coins</code> , a total mathematical function or abbreviation used by later declarations.
23	<code>(st : signing_attempt_state) : random_coins =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
24	<code>st.'1.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
25	<code>op signing_attempt_state_sample</code>	Defines operation <code>signing_attempt_state_sample</code> , a total mathematical function or abbreviation used by later declarations.
26	<code>(st : signing_attempt_state) : signing_attempt_sample =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
27	<code>st.'2.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
28	<code>op signing_attempt_sample_y1 (s : signing_attempt_sample) : polyvecl =</code>	Defines operation <code>signing_attempt_sample_y1</code> , a total mathematical function or abbreviation used by later declarations.
29	<code>s.'1.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
30	<code>op signing_attempt_sample_y2 (s : signing_attempt_sample) : polyveck =</code>	Defines operation <code>signing_attempt_sample_y2</code> , a total mathematical function or abbreviation used by later declarations.
31	<code>s.'2.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
32	<code>op signing_attempt_sample_commitment_raw</code>	Defines operation <code>signing_attempt_sample_commitment_raw</code> , a total mathematical function or abbreviation used by later declarations.
33	<code>(s : signing_attempt_sample) : polyveck =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
34	<code>s.'3.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
35	<code>op signing_attempt_state_sampled_y1</code>	Defines operation <code>signing_attempt_state_sampled_y1</code> , a total mathematical function or abbreviation used by later declarations.
36	<code>(st : signing_attempt_state) : polyvecl =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
37	<code>signing_attempt_sample_y1 (signing_attempt_state_sample st).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
38	<code>op signing_attempt_state_sampled_y2</code>	Defines operation <code>signing_attempt_state_sampled_y2</code> , a total mathematical function or abbreviation used by later declarations.
39	<code>(st : signing_attempt_state) : polyveck =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
40	<code>signing_attempt_sample_y2 (signing_attempt_state_sample st).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
41	<code>op signing_attempt_state_sampled_y</code>	Defines operation <code>signing_attempt_state_sampled_y</code> , a total mathematical function or abbreviation used by later declarations.
42	<code>(st : signing_attempt_state) : polyveck =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
43	<code>signing_attempt_sample_commitment_raw (signing_attempt_state_sample st).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
44	<code>op signing_attempt_state_highbits</code>	Defines operation <code>signing_attempt_state_highbits</code> , a total mathematical function or abbreviation used by later declarations.
45	<code>(st : signing_attempt_state) : polyveck =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
46	<code>st.'3.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
47	op signing_attempt_state_lowbits	Defines operation <code>signing_attempt_state_lowbits</code> , a total mathematical function or abbreviation used by later declarations.
48	(st : signing_attempt_state) : poly =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
49	st.'4.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
50	op signing_attempt_state_challenge	Defines operation <code>signing_attempt_state_challenge</code> , a total mathematical function or abbreviation used by later declarations.
51	(st : signing_attempt_state) : challenge =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
52	st.'5.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
53	op signing_attempt_state_response	Defines operation <code>signing_attempt_state_response</code> , a total mathematical function or abbreviation used by later declarations.
54	(st : signing_attempt_state) : polyvecl =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
55	st.'6.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
56	op signing_attempt_state_response_aux	Defines operation <code>signing_attempt_state_response_aux</code> , a total mathematical function or abbreviation used by later declarations.
57	(st : signing_attempt_state) : polyveck =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
58	st.'7.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
59	op signing_attempt_state_hint	Defines operation <code>signing_attempt_state_hint</code> , a total mathematical function or abbreviation used by later declarations.
60	(st : signing_attempt_state) : hint_t =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
61	st.'8.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
62	op signing_attempt_state_signature	Defines operation <code>signing_attempt_state_signature</code> , a total mathematical function or abbreviation used by later declarations.
63	(st : signing_attempt_state) : signature =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
64	st.'9.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
65	op signature_rejection_branch_byte (sig : signature) : int =	Defines integer-valued parameter <code>signature_rejection_branch_byte</code> , used for dimensions, budgets, support sizes, or bounds.
66	sig.'6.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
67	op signing_attempt_state_reject2_branch_byte	Defines operation <code>signing_attempt_state_reject2_branch_byte</code> , a total mathematical function or abbreviation used by later declarations.
68	(st : signing_attempt_state) : int =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
69	signature_rejection_branch_byte (signing_attempt_state_signature st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
70	op signing_attempt_state_reject2_branch_bit	Defines operation <code>signing_attempt_state_reject2_branch_bit</code> , a total mathematical function or abbreviation used by later declarations.
71	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
72	(signing_attempt_state_reject2_branch_byte st %/ 2) %% 2 = 1.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
73	op polyvecl_double (md : mode) (xs : polyvecl) : polyvecl =	Defines operation <code>polyvecl_double</code> , a total mathematical function or abbreviation used by later declarations.
74	mkseq	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
75	(fun i => poly_double (nth poly_zero xs i))	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
76	(mode_l md).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
77	op signing_attempt_state_reject2_sampled_y1	Defines operation <code>signing_attempt_state_reject2_sampled_y1</code> , a total mathematical function or abbreviation used by later declarations.
78	(st : signing_attempt_state) : polyvecl =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
79	signing_attempt_state_sampled_y1 st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
80	op signing_attempt_state_reject2_sampled_y2	Defines operation <code>signing_attempt_state_reject2_sampled_y2</code> , a total mathematical function or abbreviation used by later declarations.
81	(st : signing_attempt_state) : polyveck =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
82	signing_attempt_state_sampled_y2 st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
83	op signing_attempt_state_reject2_balance_response	Defines operation <code>signing_attempt_state_reject2_balance_response</code> , a total mathematical function or abbreviation used by later declarations.
84	(md : mode) (st : signing_attempt_state) : polyvecl =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
85	polyvecl_sub md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
86	(polyvecl_double md (signing_attempt_state_response st))	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
87	(signing_attempt_state_reject2_sampled_y1 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
88	op signing_attempt_state_reject2_balance_aux	Defines operation <code>signing_attempt_state_reject2_balance_aux</code> , a total mathematical function or abbreviation used by later declarations.
89	(md : mode) (st : signing_attempt_state) : polyveck =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
90	polyveck_sub md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
91	(polyveck_double md (signing_attempt_state_response_aux st))	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
92	(signing_attempt_state_reject2_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
93	op signing_attempt_state_lowbits_carrier	Defines operation <code>signing_attempt_state_lowbits_carrier</code> , a total mathematical function or abbreviation used by later declarations.
94	(st : signing_attempt_state) : poly =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
95	nth poly_zero (signing_attempt_state_sampled_y st) 0.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
96	op signing_attempt_state_token	Defines operation <code>signing_attempt_state_token</code> , a total mathematical function or abbreviation used by later declarations.
97	(st : signing_attempt_state) : sig_token =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
98	(signing_attempt_state_response st,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
99	signing_attempt_state_response_aux st,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
100	signing_attempt_state_hint st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
101	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
102	op signing_attempt_state_programmed_lowbits	Defines operation <code>signing_attempt_state_programmed_lowbits</code> , a total mathematical function or abbreviation used by later declarations.
103	(st : signing_attempt_state) : poly =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
104	mkseq	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
105	(fun i =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
106	if i = 0 then signing_entropy_token_of_coins	Splits proof obligations according to a conditional branch in the program or formula.
107	(signing_attempt_state_coins st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
108	else poly_coeff (signing_attempt_state_lowbits_carrier st) i)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
109	n.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
110	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
111	op signing_attempt_state_lowbits_consistent	Defines operation <code>signing_attempt_state_lowbits_consistent</code> , a total mathematical function or abbreviation used by later declarations.
112	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.

Line	EasyCrypt source	Mathematical reading
113	<code>signing_attempt_state_lowbits st =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
114	<code>signing_attempt_state_programmed_lowbits st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
115	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
116	<code>op signing_attempt_state_programmed_challenge</code>	Defines operation <code>signing_attempt_state_programmed_challenge</code> , a total mathematical function or abbreviation used by later declarations.
117	<code>(md : mode) (pk : pkey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
118	<code>(st : signing_attempt_state) : challenge =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
119	<code>challenge_hash md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
120	<code>(signing_attempt_state_response_aux st)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
121	<code>(signing_attempt_state_lowbits st)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
122	<code>(message_hash pk ctx m).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
123	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
124	<code>op signing_attempt_state_challenge_consistent</code>	Defines operation <code>signing_attempt_state_challenge_consistent</code> , a total mathematical function or abbreviation used by later declarations.
125	<code>(md : mode) (pk : pkey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
126	<code>(st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
127	<code>signing_attempt_state_challenge st =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
128	<code>signing_attempt_state_programmed_challenge md pk m ctx st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
129	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
130	<code>op signing_attempt_state_reconstructed_highbits</code>	Defines operation <code>signing_attempt_state_reconstructed_highbits</code> , a total mathematical function or abbreviation used by later declarations.
131	<code>(md : mode) (pk : pkey) (st : signing_attempt_state) : polyveck =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
132	<code>reconstructed_highbits md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
133	<code>(signing_attempt_state_response_aux st)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
134	<code>(public_key_challenge_term md pk (signing_attempt_state_challenge st)).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
135	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
136	<code>op signing_attempt_state_public_equation_consistent</code>	Defines operation <code>signing_attempt_state_public_equation_consistent</code> , a total mathematical function or abbreviation used by later declarations.
137	<code>(md : mode) (pk : pkey) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
138	<code>signing_attempt_state_highbits st =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
139	<code>signing_attempt_state_reconstructed_highbits md pk st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
140	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
141	<code>op signing_attempt_state_signature_of_fields</code>	Defines operation <code>signing_attempt_state_signature_of_fields</code> , a total mathematical function or abbreviation used by later declarations.
142	<code>(pk : pkey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
143	<code>(st : signing_attempt_state) : signature =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
144	<code>(signing_attempt_state_highbits st,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
145	<code>signing_attempt_state_lowbits st,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
146	message_hash pk ctx m,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
147	signing_attempt_state_challenge st,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
148	signing_attempt_state_token st,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
149	0,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
150	0).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
151	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
152	op signing_attempt_state_fields_match_signature	Defines operation <code>signing_attempt_state_fields_match_signature</code> , a total mathematical function or abbreviation used by later declarations.
153	(pk : pkey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
154	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
155	signing_attempt_state_signature st =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
156	signing_attempt_state_signature_of_fields pk m ctx st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
157	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
158	op signing_attempt_state_field_accepts	Defines operation <code>signing_attempt_state_field_accepts</code> , a total mathematical function or abbreviation used by later declarations.
159	(md : mode) (pk : pkey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
160	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
161	signing_attempt_state_lowbits_consistent st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
162	signing_attempt_state_challenge_consistent md pk m ctx st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
163	signing_attempt_state_public_equation_consistent md pk st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
164	signing_attempt_state_fields_match_signature pk m ctx st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
165	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
166	op signing_attempt_state_of_coins	Defines operation <code>signing_attempt_state_of_coins</code> , a total mathematical function or abbreviation used by later declarations.
167	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
168	(coins : random_coins) : signing_attempt_state =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
169	(coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
170	(signing_sample_y1 md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
171	signing_sample_y2 md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
172	commitment_raw md sk m ctx coins),	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
173	commitment_highbits md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
174	commitment_lowbits md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
175	commitment_challenge md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
176	response_vector md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
177	response_aux_vector md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
178	hint_vector md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
179	<code>sign_internal md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
180	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
181	<code>op signing_attempt_sample_of_pair</code>	Defines operation <code>signing_attempt_sample_of_pair</code> , a total mathematical function or abbreviation used by later declarations.
182	<code>(md : mode) (sk : skey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
183	<code>(coins : random_coins) (sample : signing_sample_pair) :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
184	<code>signing_attempt_sample =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
185	<code>(signing_sample_pair_y1 sample,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
186	<code>signing_sample_pair_y2 sample,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
187	<code>commitment_raw md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
188	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
189	<code>op signing_attempt_state_of_sample_pair</code>	Defines operation <code>signing_attempt_state_of_sample_pair</code> , a total mathematical function or abbreviation used by later declarations.
190	<code>(md : mode) (sk : skey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
191	<code>(coins : random_coins) (sample : signing_sample_pair) :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
192	<code>signing_attempt_state =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
193	<code>(coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
194	<code>signing_attempt_sample_of_pair md sk m ctx coins sample,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
195	<code>commitment_highbits md sk m ctx coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
196	<code>commitment_lowbits md sk m ctx coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
197	<code>commitment_challenge md sk m ctx coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
198	<code>response_vector md sk m ctx coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
199	<code>response_aux_vector md sk m ctx coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
200	<code>hint_vector md sk m ctx coins,</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
201	<code>sign_internal md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
202	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
203	<code>op signing_attempt_state_valid_signature_accepts</code>	Defines operation <code>signing_attempt_state_valid_signature_accepts</code> , a total mathematical function or abbreviation used by later declarations.
204	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
205	<code>valid_signature md (signing_attempt_state_signature st).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
206	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
207	<code>op signing_attempt_state_response_norm_accepts</code>	Defines operation <code>signing_attempt_state_response_norm_accepts</code> , a total mathematical function or abbreviation used by later declarations.
208	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
209	<code>response_norm_ok md (signing_attempt_state_signature st).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
210	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
211	<code>op signing_attempt_state_hint_norm_accepts</code>	Defines operation <code>signing_attempt_state_hint_norm_accepts</code> , a total mathematical function or abbreviation used by later declarations.

Line	EasyCrypt source	Mathematical reading
212	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
213	verify_norm_ok md (signing_attempt_state_signature st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
214	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
215	op signing_attempt_state_accepts	Defines operation signing_attempt_state_accepts, a total mathematical function or abbreviation used by later declarations.
216	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
217	signing_attempt_state_valid_signature_accepts md st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
218	signing_attempt_state_response_norm_accepts md st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
219	signing_attempt_state_hint_norm_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
220	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
221	op signing_attempt_reject1_bound (md : mode) : int =	Defines integer-valued parameter signing_attempt_reject1_bound, used for dimensions, budgets, support sizes, or bounds.
222	mode_b1sq md * mode_ln md * mode_ln md.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
223	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
224	op signing_attempt_reject2_bound (md : mode) : int =	Defines integer-valued parameter signing_attempt_reject2_bound, used for dimensions, budgets, support sizes, or bounds.
225	mode_b0sq md * mode_ln md * mode_ln md.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
226	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
227	op signing_attempt_reject2_balance_norm_sq	Defines operation signing_attempt_reject2_balance_norm_sq, a total mathematical function or abbreviation used by later declarations.
228	(md : mode) (z : polyvec1) (zaux : polyveck)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
229	(y1 : polyvec1) (y2 : polyveck) : int =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
230	polyvec1_norm_sq md (polyvec1_sub md (polyvec1_double md z) y1) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
231	polyveck_norm_sq md (polyveck_sub md (polyveck_double md zaux) y2).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
232	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
233	op signing_attempt_reject2_concrete_aborts	Defines operation signing_attempt_reject2_concrete_aborts, a total mathematical function or abbreviation used by later declarations.
234	(md : mode) (branch : bool) (z : polyvec1) (zaux : polyveck)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
235	(y1 : polyvec1) (y2 : polyveck) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
236	branch /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
237	signing_attempt_reject2_balance_norm_sq md ziaux y1 y2 <	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
238	signing_attempt_reject2_bound md.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
239	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
240	op signing_attempt_state_reject1_norm_sq	Defines operation signing_attempt_state_reject1_norm_sq, a total mathematical function or abbreviation used by later declarations.
241	(md : mode) (st : signing_attempt_state) : int =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
242	response_norm_sq md (signing_attempt_state_signature st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
243	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
244	op signing_attempt_state_reject1_accepts	Defines operation signing_attempt_state_reject1_accepts, a total mathematical function or abbreviation used by later declarations.

Line	EasyCrypt source	Mathematical reading
245	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
246	0 <= signing_attempt_state_reject1_norm_sq md st /\	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
247	signing_attempt_state_reject1_norm_sq md st <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
248	signing_attempt_reject1_bound md.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
249	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
250	op signing_attempt_state_balance_norm_sq	Defines operation signing_attempt_state_balance_norm_sq, a total mathematical function or abbreviation used by later declarations.
251	(md : mode) (st : signing_attempt_state) : int =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
252	signing_attempt_reject2_balance_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
253	(signing_attempt_state_response st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
254	(signing_attempt_state_response_aux st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
255	(signing_attempt_state_reject2_sampled_y1 st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
256	(signing_attempt_state_reject2_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
257	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
258	op signing_attempt_state_bimodal_reject2_branch	Defines operation signing_attempt_state_bimodal_reject2_branch, a total mathematical function or abbreviation used by later declarations.
259	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
260	signing_attempt_state_reject2_branch_bit st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
261	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
262	op signing_attempt_state_reject2_under_bound	Defines operation signing_attempt_state_reject2_under_bound, a total mathematical function or abbreviation used by later declarations.
263	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
264	signing_attempt_state_balance_norm_sq md st <	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
265	signing_attempt_reject2_bound md.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
266	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
267	op signing_attempt_state_reject2_aborts	Defines operation signing_attempt_state_reject2_aborts, a total mathematical function or abbreviation used by later declarations.
268	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
269	signing_attempt_reject2_concrete_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
270	(signing_attempt_state_bimodal_reject2_branch st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
271	(signing_attempt_state_response st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
272	(signing_attempt_state_response_aux st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
273	(signing_attempt_state_reject2_sampled_y1 st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
274	(signing_attempt_state_reject2_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
275	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
276	op signing_attempt_state_reject2_accepts	Defines operation signing_attempt_state_reject2_accepts, a total mathematical function or abbreviation used by later declarations.
277	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.

Line	EasyCrypt source	Mathematical reading
278	<code>! signing_attempt_state_reject2_aborts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
279	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
280	<code>op signing_attempt_state_pack_accepts</code>	Defines operation <code>signing_attempt_state_pack_accepts</code> , a total mathematical function or abbreviation used by later declarations.
281	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
282	<code>signing_attempt_state_valid_signature_accepts md st /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
283	<code>0 < mode_crypto_bytes md.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
284	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
285	<code>op signing_attempt_state_reference_rejection_accepts</code>	Defines operation <code>signing_attempt_state_reference_rejection_accepts</code> , a total mathematical function or abbreviation used by later declarations.
286	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
287	<code>signing_attempt_state_reject1_accepts md st /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
288	<code>signing_attempt_state_reject2_accepts md st /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
289	<code>signing_attempt_state_pack_accepts md st /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
290	<code>signing_attempt_state_hint_norm_accepts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
291	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
292	<code>op signing_attempt_state_reference_rejection_aborts</code>	Defines operation <code>signing_attempt_state_reference_rejection_aborts</code> , a total mathematical function or abbreviation used by later declarations.
293	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
294	<code>! signing_attempt_state_reference_rejection_accepts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
295	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
296	<code>op signing_attempt_state_reject1_aborts</code>	Defines operation <code>signing_attempt_state_reject1_aborts</code> , a total mathematical function or abbreviation used by later declarations.
297	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
298	<code>! signing_attempt_state_reject1_accepts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
299	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
300	<code>op signing_attempt_state_pack_aborts</code>	Defines operation <code>signing_attempt_state_pack_aborts</code> , a total mathematical function or abbreviation used by later declarations.
301	<code>(md : mode) (st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
302	<code>! signing_attempt_state_pack_accepts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
303	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
304	<code>op signing_attempt_state_verification_accepts</code>	Defines operation <code>signing_attempt_state_verification_accepts</code> , a total mathematical function or abbreviation used by later declarations.
305	<code>(md : mode) (pk : pkey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
306	<code>(st : signing_attempt_state) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
307	<code>signing_attempt_state_field_accepts md pk m ctx st /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
308	<code>signing_attempt_state_accepts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
309	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
310	<code>op signing_attempt_state_rejection_accepts</code>	Defines operation <code>signing_attempt_state_rejection_accepts</code> , a total mathematical function or abbreviation used by later declarations.

Line	EasyCrypt source	Mathematical reading
311	(md : mode) (pk : pkey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
312	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
313	signing_attempt_state_verification_accepts md pk m ctx st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
314	signing_attempt_state_reference_rejection_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
315	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
316	op signing_attempt_state_rejection_aborts	Defines operation signing_attempt_state_rejection_aborts, a total mathematical function or abbreviation used by later declarations.
317	(md : mode) (pk : pkey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
318	(st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
319	! signing_attempt_state_rejection_accepts md pk m ctx st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
320	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
321	op signing_attempt_state_response_norm_aborts	Defines operation signing_attempt_state_response_norm_aborts, a total mathematical function or abbreviation used by later declarations.
322	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
323	! signing_attempt_state_response_norm_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
324	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
325	op signing_attempt_state_hint_norm_aborts	Defines operation signing_attempt_state_hint_norm_aborts, a total mathematical function or abbreviation used by later declarations.
326	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
327	! signing_attempt_state_hint_norm_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
328	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
329	op signing_attempt_state_rejection_sampling_aborts	Defines operation signing_attempt_state_rejection_sampling_aborts, a total mathematical function or abbreviation used by later declarations.
330	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
331	! signing_attempt_state_valid_signature_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
332	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
333	op signing_attempt_state_aborts	Defines operation signing_attempt_state_aborts, a total mathematical function or abbreviation used by later declarations.
334	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
335	signing_attempt_state_rejection_sampling_aborts md st \/	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
336	signing_attempt_state_response_norm_aborts md st \/	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
337	signing_attempt_state_hint_norm_aborts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
338	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
339	op signing_accepts	Defines operation signing_accepts, a total mathematical function or abbreviation used by later declarations.
340	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
341	(coins : random_coins) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
342	signing_attempt_state_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
343	(signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
344	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
345	op dsigning_attempt_state	Defines operation dsigning_attempt_state, a total mathematical function or abbreviation used by later declarations.
346	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
347	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
348	dmap drandom_coins (signing_attempt_state_of_coins md sk m ctx).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
349	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
350	op signing_sample_pair_sampler_ok	Defines operation signing_sample_pair_sampler_ok, a total mathematical function or abbreviation used by later declarations.
351	(md : mode) (dsample : random_coins -> signing_sample_pair distr) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
352	bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
353	forall coins,	Part of a universal mathematical condition that must hold for all listed values.
354	signing_randomness_domain coins =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
355	signing_sample_pair_distribution_ok md (dsample coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
356	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
357	op dsigning_attempt_state_from_sample_pair_sampler	Defines operation dsigning_attempt_state_from_sample_pair_sampler, a total mathematical function or abbreviation used by later declarations.
358	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
359	(dsample : random_coins -> signing_sample_pair distr) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
360	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
361	dlet drandom_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
362	(fun coins =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
363	dmap (dsample coins)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
364	(signing_attempt_state_of_sample_pair md sk m ctx coins)).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
365	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
366	type signing_attempt_coin_sample_pair = random_coins * signing_sample_pair.	Declares carrier set signing_attempt_coin_sample_pair. EasyCrypt will type-check all later functions and games over this mathematical domain.
367	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
368	op signing_attempt_state_of_coin_sample_pair	Defines operation signing_attempt_state_of_coin_sample_pair, a total mathematical function or abbreviation used by later declarations.
369	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
370	(cs : signing_attempt_coin_sample_pair) : signing_attempt_state =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
371	signing_attempt_state_of_sample_pair md sk m ctx cs.'1 cs.'2.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
372	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
373	op dstructural_signing_attempt_coin_sample_pair	Defines operation dstructural_signing_attempt_coin_sample_pair, a total mathematical function or abbreviation used by later declarations.
374	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
375	signing_attempt_coin_sample_pair distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
376	dlet drandom_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
377	(fun coins =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
378	dmap (dstructural_signing_sample_pair md sk m ctx coins)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
379	(fun sample => (coins, sample))).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
380	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
381	op dexact_hyperball_signing_attempt_coin_sample_pair	Defines operation dexact_hyperball_signing_attempt_coin_sample_pair, a total mathematical function or abbreviation used by later declarations.
382	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
383	signing_attempt_coin_sample_pair distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
384	dlet drandom_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
385	(fun coins =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
386	dmap (dexact_hyperball_signing_sample_pair md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
387	(fun sample => (coins, sample))).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
388	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
389	op dstructural_signing_attempt_state	Defines operation dstructural_signing_attempt_state, a total mathematical function or abbreviation used by later declarations.
390	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
391	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
392	dmap drandom_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
393	(fun coins =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
394	signing_attempt_state_of_sample_pair md sk m ctx coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
395	(signing_sample_pair_of_coins md sk m ctx coins)).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
396	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
397	op dstructural_signing_attempt_state_from_sampler	Defines operation dstructural_signing_attempt_state_from_sampler, a total mathematical function or abbreviation used by later declarations.
398	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
399	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
400	dsigning_attempt_state_from_sample_pair_sampler md sk m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
401	(dstructural_signing_sample_pair md sk m ctx).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
402	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
403	op dbounded_unit_signing_attempt_state	Defines operation dbounded_unit_signing_attempt_state, a total mathematical function or abbreviation used by later declarations.
404	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
405	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
406	dsigning_attempt_state_from_sample_pair_sampler md sk m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
407	(fun _ => dbounded_unit_signing_sample_pair md).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
408	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
409	op dchecked_hyperball_signing_attempt_state	Defines operation dchecked_hyperball_signing_attempt_state, a total mathematical function or abbreviation used by later declarations.

Line	EasyCrypt source	Mathematical reading
410	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
411	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
412	dsigning_attempt_state_from_sample_pair_sampler md sk m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
413	(fun _ => dchecked_hyperball_signing_sample_pair md).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
414	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
415	op dexact_hyperball_signing_attempt_state	Defines operation dexact_hyperball_signing_attempt_state, a total mathematical function or abbreviation used by later declarations.
416	(md : mode) (sk : skey) (m : message) (ctx : context) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
417	signing_attempt_state distr =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
418	dsigning_attempt_state_from_sample_pair_sampler md sk m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
419	(fun _ => dexact_hyperball_signing_sample_pair md).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
420	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
421	op signing_attempt_state_sample_pair_wf	Defines operation signing_attempt_state_sample_pair_wf, a total mathematical function or abbreviation used by later declarations.
422	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
423	polyvecl_wf md (signing_attempt_state_sampled_y1 st) /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
424	polyveck_wf md (signing_attempt_state_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
425	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
426	op signing_attempt_state_sample_pair_unit	Defines operation signing_attempt_state_sample_pair_unit, a total mathematical function or abbreviation used by later declarations.
427	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
428	polyvecl_unit md (signing_attempt_state_sampled_y1 st) /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
429	polyveck_unit md (signing_attempt_state_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
430	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
431	op signing_attempt_state_sample_pair_bounded	Defines operation signing_attempt_state_sample_pair_bounded, a total mathematical function or abbreviation used by later declarations.
432	(md : mode) (st : signing_attempt_state) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
433	polyvecl_bounded_by md (signing_sample_bound md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
434	(signing_attempt_state_sampled_y1 st) /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
435	polyveck_bounded_by md (signing_sample_bound md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
436	(signing_attempt_state_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
437	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
438	op signing_attempt_relation	Defines operation signing_attempt_relation, a total mathematical function or abbreviation used by later declarations.
439	(md : mode) (pk : pkey) (sk : skey) (m : message)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
440	(ctx : context) (coins : random_coins) (sig : signature) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
441	valid_keypair md pk sk /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
442	signing_accepts md sk m ctx coins /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
443	<code>sig = sign_internal md sk m ctx coins.</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
444	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
445	<code>op signing_simulation_relation</code>	Defines operation <code>signing_simulation_relation</code> , a total mathematical function or abbreviation used by later declarations.
446	<code>(md : mode) (pk : pkey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
447	<code>(sig : signature) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
448	<code>verify_internal md pk m ctx sig /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
449	<code>exists (sk : skey) (coins : random_coins),</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
450	<code>signing_attempt_relation md pk sk m ctx coins sig.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
451	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
452	<code>op signing_transcript_relation</code>	Defines operation <code>signing_transcript_relation</code> , a total mathematical function or abbreviation used by later declarations.
453	<code>(md : mode) (pk : pkey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
454	<code>(sig : signature) (tr : transcript) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
455	<code>signing_simulation_relation md pk m ctx sig /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
456	<code>tr = transcript_of_signature md pk m ctx sig /\</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
457	<code>transcript_valid md tr.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
458	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
459	<code>op signing_record_message (r : signing_transcript_record) : message = r.'1.</code>	Defines operation <code>signing_record_message</code> , a total mathematical function or abbreviation used by later declarations.
460	<code>op signing_record_context (r : signing_transcript_record) : context = r.'2.</code>	Defines operation <code>signing_record_context</code> , a total mathematical function or abbreviation used by later declarations.
461	<code>op signing_record_signature (r : signing_transcript_record) : signature = r.'3.</code>	Defines operation <code>signing_record_signature</code> , a total mathematical function or abbreviation used by later declarations.
462	<code>op signing_record_transcript (r : signing_transcript_record) : transcript =</code>	Defines operation <code>signing_record_transcript</code> , a total mathematical function or abbreviation used by later declarations.
463	<code>r.'4.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
464	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
465	<code>op signing_record_query (r : signing_transcript_record) : message * context =</code>	Defines operation <code>signing_record_query</code> , a total mathematical function or abbreviation used by later declarations.
466	<code>(signing_record_message r, signing_record_context r).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
467	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
468	<code>op signing_record_relation</code>	Defines operation <code>signing_record_relation</code> , a total mathematical function or abbreviation used by later declarations.
469	<code>(md : mode) (pk : pkey) (r : signing_transcript_record) : bool =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
470	<code>signing_transcript_relation md pk</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
471	<code>(signing_record_message r)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
472	<code>(signing_record_context r)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
473	<code>(signing_record_signature r)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
474	<code>(signing_record_transcript r).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
475	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
476	op signing_record_queries (rs : signing_transcript_record list) :	Defines operation signing_record_queries, a total mathematical function or abbreviation used by later declarations.
477	(message * context) list =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
478	map signing_record_query rs.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
479	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
480	op signing_record_transcripts (rs : signing_transcript_record list) :	Defines operation signing_record_transcripts, a total mathematical function or abbreviation used by later declarations.
481	transcript list =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
482	map signing_record_transcript rs.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
483	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
484	op signing_record_log_sound	Defines operation signing_record_log_sound, a total mathematical function or abbreviation used by later declarations.
485	(md : mode) (pk : pkey) (rs : signing_transcript_record list) : bool =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
486	all (signing_record_relation md pk) rs.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
487	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
488	op transcript_log_valid (md : mode) (trs : transcript list) : bool =	Defines predicate transcript_log_valid, an event or well-formedness condition used in later implications and game proofs.
489	all (transcript_valid md) trs.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
490	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
491	op structural_to_exact_hyperball_sample_pair_point_loss_obligation : bool =	Defines predicate structural_to_exact_hyperball_sample_pair_point_loss_obligation, an event or well-formedness condition used in later implications and game proofs.
492	forall (md : mode) (sk : skey) (m : message) (ctx : context)	Part of a universal mathematical condition that must hold for all listed values.
493	(coins : random_coins),	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
494	1%r <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
495	mu (dexact_hyperball_signing_sample_pair md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
496	(pred1 (signing_sample_pair_of_coins md sk m ctx coins)) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
497	rejection_sampling_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
498	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
499	op structural_to_exact_hyperball_sample_pair_loss_obligation : bool =	Defines predicate structural_to_exact_hyperball_sample_pair_loss_obligation, an event or well-formedness condition used in later implications and game proofs.
500	forall (md : mode) (sk : skey) (m : message) (ctx : context)	Part of a universal mathematical condition that must hold for all listed values.
501	(coins : random_coins) (p : signing_sample_pair -> bool),	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
502	mu (dstructural_signing_sample_pair md sk m ctx coins) p <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
503	mu (dexact_hyperball_signing_sample_pair md) p +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
504	rejection_sampling_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
505	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
506	op structural_to_exact_hyperball_coin_sample_pair_loss_obligation : bool =	Defines predicate structural_to_exact_hyperball_coin_sample_pair_loss_obligation, an event or well-formedness condition used in later implications and game proofs.

Line	EasyCrypt source	Mathematical reading
507	forall (md : mode) (sk : skey) (m : message) (ctx : context)	Part of a universal mathematical condition that must hold for all listed values.
508	(p : signing_attempt_coin_sample_pair -> bool),	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
509	mu (dstructural_signing_attempt_coin_sample_pair md sk m ctx) p <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
510	mu (dexact_hyperball_signing_attempt_coin_sample_pair md sk m ctx) p +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
511	rejection_sampling_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
512	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
513	op structural_to_exact_hyperball_attempt_loss_obligation : bool =	Defines predicate structural_to_exact_hyperball_attempt_loss_obligation, an event or well-formedness condition used in later implications and game proofs.
514	forall (md : mode) (sk : skey) (m : message) (ctx : context)	Part of a universal mathematical condition that must hold for all listed values.
515	(p : signing_attempt_state -> bool),	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
516	mu (dsigning_attempt_state md sk m ctx) p <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
517	mu (dexact_hyperball_signing_attempt_state md sk m ctx) p +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
518	rejection_sampling_loss_term.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
519	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
520	op rejection_sampling_bound_obligation : bool =	Defines predicate rejection_sampling_bound_obligation, an event or well-formedness condition used in later implications and game proofs.
521	0%r <= rejection_sampling_loss_term /\	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
522	0%r <= fs_with_aborts_reprogramming_term /\	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
523	0%r <= fs_with_aborts_min_entropy_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
524	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
525	lemma signing_attempt_verifies md pk sk m ctx coins sig :	States checked lemma signing_attempt_verifies. The following proof script must derive this proposition from prior definitions and lemmas.
526	signing_attempt_relation md pk sk m ctx coins sig =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
527	verify_internal md pk m ctx sig.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
528	proof.	Begins the proof script for the preceding proposition.
529	rewrite /signing_attempt_relation /valid_keypair.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
530	move=> [#] pkE _ ->.	Introduces universally quantified variables or hypotheses into the proof context.
531	rewrite pkE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
532	by apply verify_internal_sign_internal.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
533	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
534	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
535	lemma signing_simulation_verifies md pk m ctx sig :	States checked lemma signing_simulation_verifies. The following proof script must derive this proposition from prior definitions and lemmas.
536	signing_simulation_relation md pk m ctx sig =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
537	verify_internal md pk m ctx sig.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
538	proof.	Begins the proof script for the preceding proposition.
539	by rewrite /signing_simulation_relation.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
540	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
541	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
542	lemma signing_attempt_simulates md pk sk m ctx coins sig :	States checked lemma signing_attempt_simulates. The following proof script must derive this proposition from prior definitions and lemmas.
543	signing_attempt_relation md pk sk m ctx coins sig =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
544	signing_simulation_relation md pk m ctx sig.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
545	proof.	Begins the proof script for the preceding proposition.
546	move=> h.	Introduces universally quantified variables or hypotheses into the proof context.
547	rewrite /signing_simulation_relation.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
548	split.	Splits a conjunction or structured goal into independent proof obligations.
549	+ by apply (signing_attempt_verifies md pk sk m ctx coins sig).	Solves a proof branch produced by a previous split, transitivity, or case analysis.
550	exists sk coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
551	by apply h.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
552	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
553	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
554	lemma sign_internal_attempt_relation md sk m ctx coins :	States checked lemma sign_internal_attempt_relation. The following proof script must derive this proposition from prior definitions and lemmas.
555	signing_attempt_relation md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
556	(public_key_of_secret md sk) sk m ctx coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
557	(sign_internal md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
558	proof.	Begins the proof script for the preceding proposition.
559	rewrite /signing_attempt_relation /valid_keypair /signing_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
560	split=> //.	Splits a conjunction or structured goal into independent proof obligations.
561	split.	Splits a conjunction or structured goal into independent proof obligations.
562	+ by rewrite /signing_attempt_state_accepts	Solves a proof branch produced by a previous split, transitivity, or case analysis.
563	/signing_attempt_state_valid_signature_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
564	/signing_attempt_state_response_norm_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
565	/signing_attempt_state_hint_norm_accepts;	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
566	smt(valid_signature_sign_internal response_norm_ok_current	Calls SMT automation for first-order arithmetic, list, or algebraic side conditions.
567	verify_norm_ok_current).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
568	+ by trivial.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
569	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
570	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
571	lemma signing_attempt_sample_of_pair_currentE md sk m ctx coins :	States checked lemma signing_attempt_sample_of_pair_currentE. The following proof script must derive this proposition from prior definitions and lemmas.
572	signing_attempt_sample_of_pair md sk m ctx coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
573	(signing_sample_pair_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
574	(signing_sample_y1 md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
575	signing_sample_y2 md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
576	commitment_raw md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
577	proof.	Begins the proof script for the preceding proposition.
578	by rewrite /signing_attempt_sample_of_pair	Discharges the current goal in one step using the cited simplification, lemma, or automation.

Line	EasyCrypt source	Mathematical reading
579	<code>/signing_sample_pair_of_coins</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
580	<code>/signing_sample_pair_y1 /signing_sample_pair_y2.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
581	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
582	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
583	<code>lemma signing_attempt_state_of_sample_pair_currentE md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_currentE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
584	<code>signing_attempt_state_of_sample_pair md sk m ctx coins</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
585	<code>(signing_sample_pair_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
586	<code>signing_attempt_state_of_coins md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
587	<code>proof.</code>	Begins the proof script for the preceding proposition.
588	<code>by rewrite /signing_attempt_state_of_sample_pair</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
589	<code>/signing_attempt_state_of_coins</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
590	<code>signing_attempt_sample_of_pair_currentE.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
591	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
592	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
593	<code>lemma structural_signing_sample_pair_sampler_ok md sk m ctx :</code>	States checked lemma <code>structural_signing_sample_pair_sampler_ok</code> . The following proof script must derive this proposition from prior definitions and lemmas.
594	<code>signing_sample_pair_sampler_ok md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
595	<code>(dstructural_signing_sample_pair md sk m ctx).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
596	<code>proof.</code>	Begins the proof script for the preceding proposition.
597	<code>rewrite /signing_sample_pair_sampler_ok => coins _.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
598	<code>by apply structural_signing_sample_pair_distribution_ok.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
599	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
600	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
601	<code>lemma bounded_unit_signing_sample_pair_sampler_ok md :</code>	States checked lemma <code>bounded_unit_signing_sample_pair_sampler_ok</code> . The following proof script must derive this proposition from prior definitions and lemmas.
602	<code>signing_sample_pair_sampler_ok md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
603	<code>(fun _ => dbounded_unit_signing_sample_pair md).</code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
604	<code>proof.</code>	Begins the proof script for the preceding proposition.
605	<code>rewrite /signing_sample_pair_sampler_ok => coins _.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
606	<code>by apply bounded_unit_signing_sample_pair_distribution_ok.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
607	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
608	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
609	<code>lemma checked_hyperball_signing_sample_pair_sampler_ok md :</code>	States checked lemma <code>checked_hyperball_signing_sample_pair_sampler_ok</code> . The following proof script must derive this proposition from prior definitions and lemmas.
610	<code>signing_sample_pair_sampler_ok md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
611	<code>(fun _ => dchecked_hyperball_signing_sample_pair md).</code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
612	<code>proof.</code>	Begins the proof script for the preceding proposition.
613	<code>rewrite /signing_sample_pair_sampler_ok => coins _.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.

Line	EasyCrypt source	Mathematical reading
614	by apply checked_hyperball_signing_sample_pair_distribution_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
615	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
616	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
617	lemma exact_hyperball_signing_sample_pair_sampler_ok md :	States checked lemma exact_hyperball_signing_sample_pair_sampler_ok. The following proof script must derive this proposition from prior definitions and lemmas.
618	signing_sample_pair_sampler_ok md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
619	(fun _ => dexact_hyperball_signing_sample_pair md).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
620	proof.	Begins the proof script for the preceding proposition.
621	rewrite /signing_sample_pair_sampler_ok => coins _.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
622	by apply exact_hyperball_signing_sample_pair_distribution_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
623	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
624	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
625	lemma dsigning_attempt_state_from_sample_pair_sampler_lossless	States checked lemma dsigning_attempt_state_from_sample_pair_sampler_lossless. The following proof script must derive this proposition from prior definitions and lemmas.
626	md sk m ctx dsample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
627	signing_sample_pair_sampler_ok md dsample =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
628	is_lossless	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
629	(dsigning_attempt_state_from_sample_pair_sampler	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
630	md sk m ctx dsample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
631	proof.	Begins the proof script for the preceding proposition.
632	move=> hsample.	Introduces universally quantified variables or hypotheses into the proof context.
633	rewrite /dsigning_attempt_state_from_sample_pair_sampler.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
634	apply dlet_ll.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
635	+ by apply signing_coin_distribution_lossless.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
636	move=> coins coins_supp.	Introduces universally quantified variables or hypotheses into the proof context.
637	apply dmap_ll.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
638	move: (hsample coins _).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
639	+ by apply signing_coin_distribution_domain.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
640	by rewrite /signing_sample_pair_distribution_ok => -[hll _].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
641	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
642	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
643	lemma dsigning_attempt_state_from_sample_pair_sampler_sample_ok	States checked lemma dsigning_attempt_state_from_sample_pair_sampler_sample_ok. The following proof script must derive this proposition from prior definitions and lemmas.
644	md sk m ctx dsample st :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
645	signing_sample_pair_sampler_ok md dsample =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
646	st \in	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
647	<code>dsigning_attempt_state_from_sample_pair_sampler</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
648	<code>md sk m ctx dsample =></code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
649	<code>signing_attempt_state_sample_pair_wf md st /\</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
650	<code>signing_attempt_state_sample_pair_bounded md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
651	<code>proof.</code>	Begins the proof script for the preceding proposition.
652	<code>move=> hsample.</code>	Introduces universally quantified variables or hypotheses into the proof context.
653	<code>rewrite /dsigning_attempt_state_from_sample_pair_sampler supp_dlet.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
654	<code>move=> [coins [coins_supp]].</code>	Introduces universally quantified variables or hypotheses into the proof context.
655	<code>rewrite supp_dmap.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
656	<code>move=> [sample [sample_supp ->]].</code>	Introduces universally quantified variables or hypotheses into the proof context.
657	<code>move: (hsample coins _).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
658	<code>+ by apply signing_coin_distribution_domain.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
659	<code>rewrite /signing_sample_pair_distribution_ok => -[_ hsupp].</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
660	<code>move: (hsupp sample sample_supp).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
661	<code>rewrite /signing_sample_pair_sample_ok => -[sample_wf sample_bounded].</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
662	<code>rewrite /signing_attempt_state_sample_pair_wf</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
663	<code>/signing_attempt_state_sample_pair_bounded</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
664	<code>/signing_attempt_state_sampled_y1</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
665	<code>/signing_attempt_state_sampled_y2</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
666	<code>/signing_attempt_sample_y1</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
667	<code>/signing_attempt_sample_y2</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
668	<code>/signing_attempt_state_sample</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
669	<code>/signing_attempt_state_of_sample_pair</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
670	<code>/signing_attempt_sample_of_pair</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
671	<code>/signing_sample_pair_wf</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
672	<code>/signing_sample_pair_bounded</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
673	<code>/signing_sample_pair_y1</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
674	<code>/signing_sample_pair_y2 /=.</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
675	<code>by split.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
676	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
677	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
678	<code>lemma dstructural_signing_attempt_state_lossless md sk m ctx :</code>	States checked lemma <code>dstructural_signing_attempt_state_lossless</code> . The following proof script must derive this proposition from prior definitions and lemmas.
679	<code>is_lossless (dstructural_signing_attempt_state md sk m ctx).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
680	<code>proof.</code>	Begins the proof script for the preceding proposition.

Line	EasyCrypt source	Mathematical reading
681	rewrite /dstructural_signing_attempt_state.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
682	by apply dmap_ll; apply signing_coin_distribution_lossless.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
683	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
684	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
685	lemma dstructural_signing_attempt_state_from_sampler_lossless	States checked lemma
686	md sk m ctx :	dstructural_signing_attempt_state_from_sampler_lossless. The following proof script must derive this proposition from prior definitions and lemmas.
687	is_lossless	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
688	(dstructural_signing_attempt_state_from_sampler md sk m ctx).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
689	proof.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
690	rewrite /dstructural_signing_attempt_state_from_sampler.	Restates the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
691	apply dsigning_attempt_state_from_sample_pair_sampler_lossless.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
692	by apply structural_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
693	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
694	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
695	lemma dbounded_unit_signing_attempt_state_lossless md sk m ctx :	States checked lemma dbounded_unit_signing_attempt_state_lossless.
696	is_lossless (dbounded_unit_signing_attempt_state md sk m ctx).	The following proof script must derive this proposition from prior definitions and lemmas.
697	proof.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
698	rewrite /dbounded_unit_signing_attempt_state.	Restates the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
699	apply dsigning_attempt_state_from_sample_pair_sampler_lossless.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
700	by apply bounded_unit_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
701	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
702	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
703	lemma dbounded_unit_signing_attempt_state_sample_ok md sk m ctx st :	States checked lemma dbounded_unit_signing_attempt_state_sample_ok.
704	st \in dbounded_unit_signing_attempt_state md sk m ctx =>	The following proof script must derive this proposition from prior definitions and lemmas.
705	signing_attempt_state_sample_pair_wf md st /\	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
706	signing_attempt_state_sample_pair_bounded md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
707	proof.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
708	rewrite /dbounded_unit_signing_attempt_state.	Restates the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
709	apply dsigning_attempt_state_from_sample_pair_sampler_sample_ok.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
710	by apply bounded_unit_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
711	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
712	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
713	lemma dchecked_hyperball_signing_attempt_state_lossless md sk m ctx :	States checked lemma dchecked_hyperball_signing_attempt_state_lossless.
714	is_lossless (dchecked_hyperball_signing_attempt_state md sk m ctx).	The following proof script must derive this proposition from prior definitions and lemmas.
715	proof.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
		Restates the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.

Line	EasyCrypt source	Mathematical reading
716	rewrite /dchecked_hyperball_signing_attempt_state.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
717	apply dsigning_attempt_state_from_sample_pair_sampler_lossless.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
718	by apply checked_hyperball_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
719	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
720	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
721	lemma dchecked_hyperball_signing_attempt_state_sample_ok md sk m ctx st :	States checked lemma dchecked_hyperball_signing_attempt_state_sample_ok. The following proof script must derive this proposition from prior definitions and lemmas.
722	st \in dchecked_hyperball_signing_attempt_state md sk m ctx =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
723	signing_attempt_state_sample_pair_wf md st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
724	signing_attempt_state_sample_pair_bounded md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
725	proof.	Begins the proof script for the preceding proposition.
726	rewrite /dchecked_hyperball_signing_attempt_state.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
727	apply dsigning_attempt_state_from_sample_pair_sampler_sample_ok.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
728	by apply checked_hyperball_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
729	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
730	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
731	lemma dexact_hyperball_signing_attempt_state_lossless md sk m ctx :	States checked lemma dexact_hyperball_signing_attempt_state_lossless. The following proof script must derive this proposition from prior definitions and lemmas.
732	is_lossless (dexact_hyperball_signing_attempt_state md sk m ctx).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
733	proof.	Begins the proof script for the preceding proposition.
734	rewrite /dexact_hyperball_signing_attempt_state.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
735	apply dsigning_attempt_state_from_sample_pair_sampler_lossless.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
736	by apply exact_hyperball_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
737	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
738	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
739	lemma dexact_hyperball_signing_attempt_state_sample_ok md sk m ctx st :	States checked lemma dexact_hyperball_signing_attempt_state_sample_ok. The following proof script must derive this proposition from prior definitions and lemmas.
740	st \in dexact_hyperball_signing_attempt_state md sk m ctx =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
741	signing_attempt_state_sample_pair_wf md st /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
742	signing_attempt_state_sample_pair_bounded md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
743	proof.	Begins the proof script for the preceding proposition.
744	rewrite /dexact_hyperball_signing_attempt_state.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
745	apply dsigning_attempt_state_from_sample_pair_sampler_sample_ok.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
746	by apply exact_hyperball_signing_sample_pair_sampler_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
747	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
748	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
749	lemma dexact_hyperball_signing_attempt_state_supportP md sk m ctx st :	States checked lemma dexact_hyperball_signing_attempt_state_supportP. The following proof script must derive this proposition from prior definitions and lemmas.
750	st \in dexact_hyperball_signing_attempt_state md sk m ctx <=>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.

Line	EasyCrypt source	Mathematical reading
751	exists (coins : random_coins) (sample : signing_sample_pair),	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
752	coins \in drandom_coins /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
753	hyperball_sample_ok md sample /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
754	st = signing_attempt_state_of_sample_pair md sk m ctx coins sample.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
755	proof.	Begins the proof script for the preceding proposition.
756	rewrite /dexact_hyperball_signing_attempt_state	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
757	/dsigning_attempt_state_from_sample_pair_sampler supp_dlet.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
758	split.	Splits a conjunction or structured goal into independent proof obligations.
759	+ move=> [coins [coins_supp]].	Solves a proof branch produced by a previous split, transitivity, or case analysis.
760	rewrite supp_dmap.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
761	move=> [sample [sample_supp ->]].	Introduces universally quantified variables or hypotheses into the proof context.
762	exists coins sample.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
763	split=> //.	Splits a conjunction or structured goal into independent proof obligations.
764	split.	Splits a conjunction or structured goal into independent proof obligations.
765	+ by move: sample_supp; rewrite exact_hyperball_signing_sample_pair_supportP.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
766	by [].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
767	move=> [coins sample [coins_supp [sample_ok stE]]].	Introduces universally quantified variables or hypotheses into the proof context.
768	rewrite stE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
769	exists coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
770	split=> //.	Splits a conjunction or structured goal into independent proof obligations.
771	rewrite supp_dmap.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
772	exists sample.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
773	split.	Splits a conjunction or structured goal into independent proof obligations.
774	+ by rewrite exact_hyperball_signing_sample_pair_supportP.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
775	by [].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
776	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
777	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
778	lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_bound1	States checked lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_bound1. The following proof script must derive this proposition from prior definitions and lemmas.
779	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
780	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
781	(signing_attempt_state_reference_rejection_aborts md) <= 1/r.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
782	proof.	Begins the proof script for the preceding proposition.
783	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
784	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
785	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
786	lemma dexact_hyperball_signing_attempt_state_mu_pullback	States checked lemma dexact_hyperball_signing_attempt_state_mu_pullback. The following proof script must derive this proposition from prior definitions and lemmas.

Line	EasyCrypt source	Mathematical reading
787	md sk m ctx (P : signing_attempt_state -> bool) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
788	mu (dexact_hyperball_signing_attempt_state md sk m ctx) P =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
789	mu (dlet drandom_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
790	(fun coins =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
791	dmap (dexact_hyperball_signing_sample_pair md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
792	(fun sample =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
793	signing_attempt_state_of_sample_pair	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
794	md sk m ctx coins sample)))	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
795	P.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
796	proof.	Begins the proof script for the preceding proposition.
797	by rewrite /dexact_hyperball_signing_attempt_state	Discharges the current goal in one step using the cited simplification, lemma, or automation.
798	/dsigning_attempt_state_from_sample_pair_sampler.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
799	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
800	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
801	lemma signing_attempt_state_reference_rejection_aborts_componentE md st :	States checked lemma signing_attempt_state_reference_rejection_aborts_componentE. The following proof script must derive this proposition from prior definitions and lemmas.
802	signing_attempt_state_reference_rejection_aborts md st =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
803	(signing_attempt_state_reject1_aborts md st \/	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
804	signing_attempt_state_reject2_aborts md st \/	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
805	signing_attempt_state_pack_aborts md st \/	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
806	signing_attempt_state_hint_norm_aborts md st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
807	proof.	Begins the proof script for the preceding proposition.
808	rewrite /signing_attempt_state_reference_rejection_aborts	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
809	/signing_attempt_state_reference_rejection_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
810	/signing_attempt_state_reject1_aborts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
811	/signing_attempt_state_reject2_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
812	/signing_attempt_state_pack_aborts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
813	/signing_attempt_state_hint_norm_aborts.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
814	by smt().	Discharges the current goal in one step using the cited simplification, lemma, or automation.
815	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
816	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
817	lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_split	States checked lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_split. The following proof script must derive this proposition from prior definitions and lemmas.
818	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
819	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
820	(signing_attempt_state_reference_rejection_aborts md) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
821	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
822	(predU	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
823	(signing_attempt_state_reject1_aborts md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
824	(predU	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
825	(signing_attempt_state_reject2_aborts md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
826	(predU	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
827	(signing_attempt_state_pack_aborts md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
828	(signing_attempt_state_hint_norm_aborts md))).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
829	proof.	Begins the proof script for the preceding proposition.
830	apply mu_eq => st.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
831	by rewrite signing_attempt_state_reference_rejection_aborts_componentE /predU.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
832	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
833	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
834	lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_union_bound	States checked lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_union_bound. The following proof script must derive this proposition from prior definitions and lemmas.
835	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
836	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
837	(signing_attempt_state_reference_rejection_aborts md) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
838	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
839	(signing_attempt_state_reject1_aborts md) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
840	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
841	(signing_attempt_state_reject2_aborts md) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
842	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
843	(signing_attempt_state_pack_aborts md) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
844	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
845	(signing_attempt_state_hint_norm_aborts md).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
846	proof.	Begins the proof script for the preceding proposition.
847	rewrite dexact_hyperball_signing_attempt_state_reference_rejection_abort_split.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
848	rewrite !mu_or.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
849	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
850	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
851	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
852	lemma signing_attempt_state_of_sample_pair_signatureE	States checked lemma signing_attempt_state_of_sample_pair_signatureE. The following proof script must derive this proposition from prior definitions and lemmas.
853	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
854	<code>signing_attempt_state_signature</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
855	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
856	<code>sign_internal md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
857	<code>proof.</code>	Begins the proof script for the preceding proposition.
858	<code>by rewrite /signing_attempt_state_signature</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
859	<code>/signing_attempt_state_of_sample_pair.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
860	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
861	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
862	<code>lemma signing_attempt_state_of_sample_pair_pack_accepts</code>	States checked lemma
863	<code>md sk m ctx coins sample :</code>	<code>signing_attempt_state_of_sample_pair_pack_accepts</code> . The following proof script must derive this proposition from prior definitions and lemmas.
864	<code>signing_attempt_state_pack_accepts md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
865	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
866	<code>proof.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
867	<code>rewrite /signing_attempt_state_pack_accepts.</code>	Begins the proof script for the preceding proposition.
868	<code>split.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
869	<code>+ by rewrite /signing_attempt_state_valid_signature_accepts</code>	Splits a conjunction or structured goal into independent proof obligations.
870	<code>signing_attempt_state_of_sample_pair_signatureE</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
871	<code>valid_signature_sign_internal.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
872	<code>by apply mode_crypto_bytes_gt0.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
873	<code>qed.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
874	<code>blank</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
875	<code>lemma signing_attempt_state_of_sample_pair_pack_no_abort</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
876	<code>md sk m ctx coins sample :</code>	States checked lemma
877	<code>! signing_attempt_state_pack_aborts md</code>	<code>signing_attempt_state_of_sample_pair_pack_no_abort</code> . The following proof script must derive this proposition from prior definitions and lemmas.
878	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
879	<code>proof.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
880	<code>by rewrite /signing_attempt_state_pack_aborts</code>	Begins the proof script for the preceding proposition.
881	<code>signing_attempt_state_of_sample_pair_pack_accepts.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
882	<code>qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
883	<code>blank</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
884	<code>lemma dexact_hyperball_signing_attempt_state_pack_abort_zero</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
885	<code>md sk m ctx :</code>	States checked lemma
886	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	<code>dexact_hyperball_signing_attempt_state_pack_abort_zero</code> . The following proof script must derive this proposition from prior definitions and lemmas.
887	<code>(signing_attempt_state_pack_aborts md) =</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
888	<code>0%r.</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
		Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
889	<code>proof.</code>	Begins the proof script for the preceding proposition.
890	<code>have le0 :</code>	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
891	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
892	<code>(signing_attempt_state_pack_aborts md) <=</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
893	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx) pred0.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
894	<code>+ apply mu_le => st st_supp abort.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
895	<code>move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
896	<code>move=> [coins sample [_ [_ stE]]].</code>	Introduces universally quantified variables or hypotheses into the proof context.
897	<code>move: abort.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
898	<code>rewrite stE signing_attempt_state_of_sample_pair_pack_no_abort.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
899	<code>by [].</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
900	<code>rewrite mu0 in le0.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
901	<code>by smt(mu_bounded).</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
902	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
903	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
904	<code>lemma dstructural_signing_attempt_state_from_samplerE md sk m ctx :</code>	States checked lemma <code>dstructural_signing_attempt_state_from_samplerE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
905	<code>dstructural_signing_attempt_state_from_sampler md sk m ctx =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
906	<code>dstructural_signing_attempt_state md sk m ctx.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
907	<code>proof.</code>	Begins the proof script for the preceding proposition.
908	<code>rewrite /dstructural_signing_attempt_state_from_sampler</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
909	<code>/dsigning_attempt_state_from_sample_pair_sampler</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
910	<code>/dstructural_signing_attempt_state</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
911	<code>/dstructural_signing_sample_pair.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
912	<code>have -> :</code>	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
913	<code>(fun coins =></code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
914	<code>dmap (dunit (signing_sample_pair_of_coins md sk m ctx coins))</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
915	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins)) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
916	<code>(dunit \o</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
917	<code>(fun coins =></code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
918	<code>signing_attempt_state_of_sample_pair md sk m ctx coins</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
919	<code>(signing_sample_pair_of_coins md sk m ctx coins))).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
920	<code>+ apply fun_ext => coins.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
921	<code>by rewrite /(\o) dmap_dunit.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
922	<code>by rewrite dlet_dunit.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.

Line	EasyCrypt source	Mathematical reading
923	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
924	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
925	<code>lemma dstructural_signing_attempt_stateE md sk m ctx :</code>	States checked lemma <code>dstructural_signing_attempt_stateE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
926	<code>dstructural_signing_attempt_state md sk m ctx =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
927	<code>dsigning_attempt_state md sk m ctx.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
928	<i>proof.</i>	Begins the proof script for the preceding proposition.
929	<code>rewrite /dstructural_signing_attempt_state /dsigning_attempt_state.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
930	<code>apply eq_dmap => coins /.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
931	<code>by rewrite signing_attempt_state_of_sample_pair_currentE.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
932	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
933	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
934	<code>lemma dstructural_signing_attempt_state_from_sampler_coin_sample_pairE</code>	States checked lemma <code>dstructural_signing_attempt_state_from_sampler_coin_sample_pairE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
935	<code>md sk m ctx :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
936	<code>dmap (dstructural_signing_attempt_coin_sample_pair md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
937	<code>(signing_attempt_state_of_coin_sample_pair md sk m ctx) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
938	<code>dstructural_signing_attempt_state_from_sampler md sk m ctx.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
939	<i>proof.</i>	Begins the proof script for the preceding proposition.
940	<code>rewrite /dstructural_signing_attempt_coin_sample_pair</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
941	<code>/dstructural_signing_attempt_state_from_sampler</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
942	<code>/dsigning_attempt_state_from_sample_pair_sampler.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
943	<code>rewrite dmap_dlet.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
944	<code>apply eq_dlet => // coins.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
945	<code>rewrite dmap_comp.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
946	<code>apply eq_dmap => sample /.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
947	<code>by rewrite /(\o) /signing_attempt_state_of_coin_sample_pair.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
948	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
949	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
950	<code>lemma dexact_hyperball_signing_attempt_state_coin_sample_pairE</code>	States checked lemma <code>dexact_hyperball_signing_attempt_state_coin_sample_pairE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
951	<code>md sk m ctx :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
952	<code>dmap (dexact_hyperball_signing_attempt_coin_sample_pair md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
953	<code>(signing_attempt_state_of_coin_sample_pair md sk m ctx) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
954	<code>dexact_hyperball_signing_attempt_state md sk m ctx.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
955	<i>proof.</i>	Begins the proof script for the preceding proposition.
956	<code>rewrite /dexact_hyperball_signing_attempt_coin_sample_pair</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.

Line	EasyCrypt source	Mathematical reading
957	/dexact_hyperball_signing_attempt_state	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
958	/dsigning_attempt_state_from_sample_pair_sampler.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
959	rewrite dmap_dlet.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
960	apply eq_dlet => // coins.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
961	rewrite dmap_comp.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
962	apply eq_dmap => sample /=.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
963	by rewrite /(\o) /signing_attempt_state_of_coin_sample_pair.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
964	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
965	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
966	lemma mu_dlet_le_add_all ['a 'b 'c]	States checked lemma mu_dlet_le_add_all. The following proof script must derive this proposition from prior definitions and lemmas.
967	(d : 'a distr) (F1 : 'a -> 'b distr) (F2 : 'a -> 'c distr)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
968	(P1 : 'b -> bool) (P2 : 'c -> bool) (eps : real) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
969	0%r <= eps =>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
970	(forall x, mu (F1 x) P1 <= mu (F2 x) P2 + eps) =>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
971	mu (dlet d F1) P1 <= mu (dlet d F2) P2 + eps.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
972	proof.	Begins the proof script for the preceding proposition.
973	move=> eps_ge h.	Introduces universally quantified variables or hypotheses into the proof context.
974	rewrite !dletE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
975	have s1_sbl : summable (fun x => mu1 d x * mu (F1 x) P1).	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
976	+ by apply summable_mu1_wght => x; smt(mu_bounded).	Solves a proof branch produced by a previous split, transitivity, or case analysis.
977	have s2_sbl : summable (fun x => mu1 d x * mu (F2 x) P2).	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
978	+ by apply summable_mu1_wght => x; smt(mu_bounded).	Solves a proof branch produced by a previous split, transitivity, or case analysis.
979	have seps_sbl : summable (fun x => eps * mu1 d x).	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
980	+ by apply summableZ; apply summable_mu1.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
981	have s2eps_sbl :	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
982	summable (fun x => mu1 d x * (mu (F2 x) P2 + eps)).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
983	+ have -> :	Solves a proof branch produced by a previous split, transitivity, or case analysis.
984	(fun x => mu1 d x * (mu (F2 x) P2 + eps)) =	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
985	(fun x => mu1 d x * mu (F2 x) P2 + eps * mu1 d x).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
986	+ by apply/fun_ext=> x; smt.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
987	by apply summableD.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
988	have hsum :	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
989	sum (fun x => mu1 d x * mu (F1 x) P1) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
990	sum (fun x => mu1 d x * (mu (F2 x) P2 + eps)).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.

Line	EasyCrypt source	Mathematical reading
991	+ apply RealSeries.ler_sum.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
992	+ move=> x /=.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
993	by apply ler_wpmul2l; [apply ge0_mu1 apply h].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
994	+ exact s1_sbl.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
995	+ exact s2eps_sbl.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
996	apply (ler_trans _ _ _ hsum).	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
997	rewrite (eq_sum _ (fun x => mu1 d x * mu (F2 x) P2 + eps * mu1 d x)).	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
998	+ by move=> x /=; smt.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
999	rewrite sumD // sumZ -weightE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1000	apply ler_add2l.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1001	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1002	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1003	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1004	lemma structural_to_exact_hyperball_sample_pair_loss_from_point_loss :	States checked lemma <u>structural_to_exact_hyperball_sample_pair_loss_from_point_loss</u> . The following proof script must derive this proposition from prior definitions and lemmas.
1005	structural_to_exact_hyperball_sample_pair_point_loss_obligation =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1006	structural_to_exact_hyperball_sample_pair_loss_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1007	proof.	Begins the proof script for the preceding proposition.
1008	rewrite /structural_to_exact_hyperball_sample_pair_point_loss_obligation	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1009	/structural_to_exact_hyperball_sample_pair_loss_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1010	move=> hop md sk m ctx coins p.	Introduces universally quantified variables or hypotheses into the proof context.
1011	rewrite /dstructural_signing_sample_pair_dunitE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1012	case: (p (signing_sample_pair_of_coins md sk m ctx coins)) => hs /=.	Performs case analysis or induction, splitting the mathematical proof by constructors or list structure.
1013	+ have hpoint := hop md sk m ctx coins.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1014	have hsub :	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1015	mu (dexact_hyperball_signing_sample_pair md)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1016	(pred1 (signing_sample_pair_of_coins md sk m ctx coins)) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1017	mu (dexact_hyperball_signing_sample_pair md) p.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1018	+ by apply mu_sub => x; smt.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1019	by smt.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1020	+ smt(mu_bounded rejection_sampling_loss_term_nonnegative).	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1021	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1022	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1023	lemma structural_to_exact_hyperball_coin_sample_pair_loss_from_sample_pair_loss :	States checked lemma <u>structural_to_exact_hyperball_coin_sample_pair_loss_from_sample_pair_loss</u> . The following proof script must derive this proposition from prior definitions and lemmas.

Line	EasyCrypt source	Mathematical reading
1024	structural_to_exact_hyperball_sample_pair_loss_obligation =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1025	structural_to_exact_hyperball_coin_sample_pair_loss_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1026	proof.	Begins the proof script for the preceding proposition.
1027	rewrite /structural_to_exact_hyperball_sample_pair_loss_obligation	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1028	/structural_to_exact_hyperball_coin_sample_pair_loss_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1029	move=> hop md sk m ctx p.	Introduces universally quantified variables or hypotheses into the proof context.
1030	rewrite /dstructural_signing_attempt_coin_sample_pair	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1031	/dexact_hyperball_signing_attempt_coin_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1032	apply mu_dlet_le_add_all.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1033	+ apply rejection_sampling_loss_term_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1034	move=> coins.	Introduces universally quantified variables or hypotheses into the proof context.
1035	rewrite !dmapE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1036	exact (hop md sk m ctx coins (fun sample => p (coins, sample))).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1037	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1038	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1039	lemma structural_to_exact_hyperball_attempt_loss_from_coin_sample_pair_loss :	States checked lemma structural_to_exact_hyperball_attempt_loss_from_coin_sample_pair_loss. The following proof script must derive this proposition from prior definitions and lemmas.
1040	structural_to_exact_hyperball_coin_sample_pair_loss_obligation =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1041	structural_to_exact_hyperball_attempt_loss_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1042	proof.	Begins the proof script for the preceding proposition.
1043	rewrite /structural_to_exact_hyperball_coin_sample_pair_loss_obligation	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1044	/structural_to_exact_hyperball_attempt_loss_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1045	move=> hop md sk m ctx p.	Introduces universally quantified variables or hypotheses into the proof context.
1046	rewrite -dstructural_signing_attempt_stateE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1047	rewrite -dstructural_signing_attempt_state_from_samplerE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1048	rewrite -dstructural_signing_attempt_state_from_sampler_coin_sample_pairE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1049	rewrite -dexact_hyperball_signing_attempt_state_coin_sample_pairE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1050	rewrite !dmapE.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1051	exact (hop md sk m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1052	(fun cs => p (signing_attempt_state_of_coin_sample_pair md sk m ctx cs))).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1053	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1054	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1055	lemma signing_attempt_state_of_coins_signatureE md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_signatureE. The following proof script must derive this proposition from prior definitions and lemmas.
1056	signing_attempt_state_signature	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1057	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.

Line	EasyCrypt source	Mathematical reading
1058	<code>sign_internal md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1059	<code>proof. by rewrite /signing_attempt_state_signature</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1060	<code>/signing_attempt_state_of_coins. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1061	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1062	<code>lemma signing_attempt_state_of_coins_sampled_y1E md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_sampled_y1E</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1063	<code>signing_attempt_state_sampled_y1</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1064	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1065	<code>signing_sample_y1 md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1066	<code>proof.</code>	Begins the proof script for the preceding proposition.
1067	<code>by rewrite /signing_attempt_state_sampled_y1</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1068	<code>/signing_attempt_sample_y1</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1069	<code>/signing_attempt_state_sample</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1070	<code>/signing_attempt_state_of_coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1071	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1072	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1073	<code>lemma signing_attempt_state_of_coins_sampled_y2E md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_sampled_y2E</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1074	<code>signing_attempt_state_sampled_y2</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1075	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1076	<code>signing_sample_y2 md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1077	<code>proof.</code>	Begins the proof script for the preceding proposition.
1078	<code>by rewrite /signing_attempt_state_sampled_y2</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1079	<code>/signing_attempt_sample_y2</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1080	<code>/signing_attempt_state_sample</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1081	<code>/signing_attempt_state_of_coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1082	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1083	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1084	<code>lemma signing_attempt_state_of_coins_sampled_yE md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_sampled_yE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1085	<code>signing_attempt_state_sampled_y</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1086	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1087	<code>commitment_raw md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1088	<code>proof.</code>	Begins the proof script for the preceding proposition.
1089	<code>by rewrite /signing_attempt_state_sampled_y</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1090	<code>/signing_attempt_sample_commitment_raw</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1091	<code>/signing_attempt_state_sample</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1092	<code>/signing_attempt_state_of_coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1093	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1094	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1095	<code>lemma signing_attempt_state_of_coins_highbitsE md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_highbitsE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1096	<code>signing_attempt_state_highbits</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1097	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1098	<code>commitment_highbits md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1099	<code>proof. by rewrite /signing_attempt_state_highbits</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1100	<code>/signing_attempt_state_of_coins. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1101	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1102	<code>lemma signing_attempt_state_of_coins_lowbitsE md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_lowbitsE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1103	<code>signing_attempt_state_lowbits</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1104	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1105	<code>commitment_lowbits md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1106	<code>proof. by rewrite /signing_attempt_state_lowbits</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1107	<code>/signing_attempt_state_of_coins. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1108	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1109	<code>lemma signing_attempt_state_of_coins_challengeE md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_challengeE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1110	<code>signing_attempt_state_challenge</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1111	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1112	<code>commitment_challenge md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1113	<code>proof. by rewrite /signing_attempt_state_challenge</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1114	<code>/signing_attempt_state_of_coins. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1115	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1116	<code>lemma signing_attempt_state_of_coins_responseE md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_responseE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1117	<code>signing_attempt_state_response</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1118	<code>(signing_attempt_state_of_coins md sk m ctx coins) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1119	<code>response_vector md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1120	<code>proof. by rewrite /signing_attempt_state_response</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1121	<code>/signing_attempt_state_of_coins. qed.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1122	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
1123	lemma signing_attempt_state_of_coins_response_auxE md sk m ctx coins :	States checked lemma <code>signing_attempt_state_of_coins_response_auxE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1124	signing_attempt_state_response_aux	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1125	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1126	response_aux_vector md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1127	proof. by rewrite /signing_attempt_state_response_aux	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1128	/signing_attempt_state_of_coins. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1129	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1130	lemma signing_attempt_state_of_coins_hintE md sk m ctx coins :	States checked lemma <code>signing_attempt_state_of_coins_hintE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1131	signing_attempt_state_hint	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1132	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1133	hint_vector md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1134	proof. by rewrite /signing_attempt_state_hint	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1135	/signing_attempt_state_of_coins. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1136	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1137	lemma signing_attempt_state_of_coins_lowbits_carrierE md sk m ctx coins :	States checked lemma <code>signing_attempt_state_of_coins_lowbits_carrierE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1138	signing_attempt_state_lowbits_carrier	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1139	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1140	nth poly_zero (commitment_raw md sk m ctx coins) 0.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1141	proof.	Begins the proof script for the preceding proposition.
1142	by rewrite /signing_attempt_state_lowbits_carrier	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1143	signing_attempt_state_of_coins_sampled_yE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1144	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1145	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1146	lemma signing_attempt_state_of_coins_tokenE md sk m ctx coins :	States checked lemma <code>signing_attempt_state_of_coins_tokenE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1147	signing_attempt_state_token	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1148	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1149	signature_token md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1150	proof.	Begins the proof script for the preceding proposition.
1151	by rewrite /signing_attempt_state_token	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1152	signing_attempt_state_of_coins_responseE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1153	signing_attempt_state_of_coins_response_auxE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1154	signing_attempt_state_of_coins_hintE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1155	/signature_token.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1156	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1157	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1158	lemma signing_attempt_state_of_coins_programmed_lowbitsE md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_programmed_lowbitsE. The following proof script must derive this proposition from prior definitions and lemmas.
1159	signing_attempt_state_programmed_lowbits	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1160	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1161	commitment_lowbits md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1162	proof.	Begins the proof script for the preceding proposition.
1163	by rewrite /signing_attempt_state_programmed_lowbits	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1164	signing_attempt_state_of_coins_lowbits_carrierE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1165	/signing_attempt_state_coins /signing_attempt_state_of_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1166	/commitment_lowbits.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1167	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1168	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1169	lemma signing_attempt_state_of_coins_lowbits_consistent md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_lowbits_consistent. The following proof script must derive this proposition from prior definitions and lemmas.
1170	signing_attempt_state_lowbits_consistent	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1171	(signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1172	proof.	Begins the proof script for the preceding proposition.
1173	by rewrite /signing_attempt_state_lowbits_consistent	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1174	signing_attempt_state_of_coins_lowbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1175	signing_attempt_state_of_coins_programmed_lowbitsE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1176	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1177	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1178	lemma signing_attempt_state_of_coins_programmed_challengeE md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_programmed_challengeE. The following proof script must derive this proposition from prior definitions and lemmas.
1179	signing_attempt_state_programmed_challenge md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1180	m ctx (signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1181	commitment_challenge md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1182	proof.	Begins the proof script for the preceding proposition.
1183	by rewrite /signing_attempt_state_programmed_challenge	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1184	signing_attempt_state_of_coins_response_auxE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1185	signing_attempt_state_of_coins_lowbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1186	/commitment_challenge.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1187	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1188	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1189	lemma signing_attempt_state_of_coins_challenge_consistent md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_challenge_consistent. The following proof script must derive this proposition from prior definitions and lemmas.
1190	signing_attempt_state_challenge_consistent md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1191	m ctx (signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1192	proof.	Begins the proof script for the preceding proposition.
1193	by rewrite /signing_attempt_state_challenge_consistent	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1194	signing_attempt_state_of_coins_challengeE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1195	signing_attempt_state_of_coins_programmed_challengeE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1196	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1197	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1198	lemma signing_attempt_state_of_coins_reconstructed_highbitsE	States checked lemma signing_attempt_state_of_coins_reconstructed_highbitsE. The following proof script must derive this proposition from prior definitions and lemmas.
1199	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1200	signing_attempt_state_reconstructed_highbits md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1201	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1202	commitment_highbits md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1203	proof.	Begins the proof script for the preceding proposition.
1204	by rewrite /signing_attempt_state_reconstructed_highbits	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1205	signing_attempt_state_of_coins_response_auxE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1206	signing_attempt_state_of_coins_challengeE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1207	/commitment_highbits.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1208	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1209	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1210	lemma signing_attempt_state_of_coins_public_equation_consistent	States checked lemma signing_attempt_state_of_coins_public_equation_consistent. The following proof script must derive this proposition from prior definitions and lemmas.
1211	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1212	signing_attempt_state_public_equation_consistent md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1213	(public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1214	(signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1215	proof.	Begins the proof script for the preceding proposition.
1216	by rewrite /signing_attempt_state_public_equation_consistent	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1217	signing_attempt_state_of_coins_highbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1218	signing_attempt_state_of_coins_reconstructed_highbitsE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1219	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1220	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1221	lemma signing_attempt_state_of_coins_signature_of_fieldsE md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_signature_of_fieldsE. The following proof script must derive this proposition from prior definitions and lemmas.
1222	signing_attempt_state_signature_of_fields	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1223	(public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1224	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.

Line	EasyCrypt source	Mathematical reading
1225	<code>sign_internal md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1226	<code>proof.</code>	Begins the proof script for the preceding proposition.
1227	<code>by rewrite /signing_attempt_state_signature_of_fields /sign_internal</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1228	<code>signing_attempt_state_of_coins_highbitsE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1229	<code>signing_attempt_state_of_coins_lowbitsE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1230	<code>signing_attempt_state_of_coins_challengeE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1231	<code>signing_attempt_state_of_coins_tokenE.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1232	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1233	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1234	<code>lemma signing_attempt_state_of_coins_fields_match_signature md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_fields_match_signature</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1235	<code>signing_attempt_state_fields_match_signature</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1236	<code>(public_key_of_secret md sk) m ctx</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1237	<code>(signing_attempt_state_of_coins md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1238	<code>proof.</code>	Begins the proof script for the preceding proposition.
1239	<code>by rewrite /signing_attempt_state_fields_match_signature</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1240	<code>signing_attempt_state_of_coins_signatureE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1241	<code>signing_attempt_state_of_coins_signature_of_fieldsE.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1242	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1243	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1244	<code>lemma signing_attempt_state_of_coins_field_accepts md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_field_accepts</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1245	<code>signing_attempt_state_field_accepts md (public_key_of_secret md sk)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1246	<code>m ctx (signing_attempt_state_of_coins md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1247	<code>proof.</code>	Begins the proof script for the preceding proposition.
1248	<code>rewrite /signing_attempt_state_field_accepts.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1249	<code>split.</code>	Splits a conjunction or structured goal into independent proof obligations.
1250	<code>+ by apply signing_attempt_state_of_coins_lowbits_consistent.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1251	<code>split.</code>	Splits a conjunction or structured goal into independent proof obligations.
1252	<code>+ by apply signing_attempt_state_of_coins_challenge_consistent.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1253	<code>split.</code>	Splits a conjunction or structured goal into independent proof obligations.
1254	<code>+ by apply signing_attempt_state_of_coins_public_equation_consistent.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1255	<code>by apply signing_attempt_state_of_coins_fields_match_signature.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1256	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1257	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1258	<code>lemma signing_attempt_state_of_coins_accepts_current md sk m ctx coins :</code>	States checked lemma <code>signing_attempt_state_of_coins_accepts_current</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1259	<code>signing_attempt_state_accepts md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1260	<code>(signing_attempt_state_of_coins md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1261	proof.	Begins the proof script for the preceding proposition.
1262	by rewrite /signing_attempt_state_accepts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1263	/signing_attempt_state_valid_signature_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1264	/signing_attempt_state_response_norm_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1265	/signing_attempt_state_hint_norm_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1266	signing_attempt_state_of_coins_signatureE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1267	valid_signature_sign_internal	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1268	response_norm_ok_current	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1269	verify_norm_ok_current.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1270	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1271	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1272	lemma signing_attempt_reject1_bound_ge_response_bound md :	States checked lemma signing_attempt_reject1_bound_ge_response_bound. The following proof script must derive this proposition from prior definitions and lemmas.
1273	mode_blsq md <= signing_attempt_reject1_bound md.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1274	proof. by case md. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1275	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1276	lemma signing_attempt_state_reject1_accepts_from_response_norm_ok md st :	States checked lemma signing_attempt_state_reject1_accepts_from_response_norm_ok. The following proof script must derive this proposition from prior definitions and lemmas.
1277	signing_attempt_state_response_norm_accepts md st =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1278	signing_attempt_state_reject1_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1279	proof.	Begins the proof script for the preceding proposition.
1280	rewrite /signing_attempt_state_response_norm_accepts	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1281	/response_norm_ok /response_norm_sq	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1282	/signing_attempt_state_reject1_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1283	/signing_attempt_state_reject1_norm_sq.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1284	move=> [norm_ge0 norm_le].	Introduces universally quantified variables or hypotheses into the proof context.
1285	split=> //.	Splits a conjunction or structured goal into independent proof obligations.
1286	by smt(signing_attempt_reject1_bound_ge_response_bound).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1287	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1288	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1289	lemma signing_attempt_state_balance_norm_sq_concreteE md st :	States checked lemma signing_attempt_state_balance_norm_sq_concreteE. The following proof script must derive this proposition from prior definitions and lemmas.
1290	signing_attempt_state_balance_norm_sq md st =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1291	signing_attempt_reject2_balance_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1292	(signing_attempt_state_response st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1293	(signing_attempt_state_response_aux st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1294	(signing_attempt_state_reject2_sampled_y1 st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1295	(signing_attempt_state_reject2_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1296	proof. by rewrite /signing_attempt_state_balance_norm_sq. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1297	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1298	lemma signing_attempt_state_reject2_under_bound_concreteE md st :	States checked lemma signing_attempt_state_reject2_under_bound_concreteE. The following proof script must derive this proposition from prior definitions and lemmas.
1299	signing_attempt_state_reject2_under_bound md st =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1300	(signing_attempt_reject2_balance_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1301	(signing_attempt_state_response st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1302	(signing_attempt_state_response_aux st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1303	(signing_attempt_state_reject2_sampled_y1 st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1304	(signing_attempt_state_reject2_sampled_y2 st) <	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1305	signing_attempt_reject2_bound md).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1306	proof.	Begins the proof script for the preceding proposition.
1307	by rewrite /signing_attempt_state_reject2_under_bound	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1308	signing_attempt_state_balance_norm_sq_concreteE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1309	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1310	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1311	lemma signing_attempt_state_reject2_aborts_concreteE md st :	States checked lemma signing_attempt_state_reject2_aborts_concreteE. The following proof script must derive this proposition from prior definitions and lemmas.
1312	signing_attempt_state_reject2_aborts md st =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1313	signing_attempt_reject2_concrete_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1314	(signing_attempt_state_bimodal_reject2_branch st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1315	(signing_attempt_state_response st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1316	(signing_attempt_state_response_aux st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1317	(signing_attempt_state_reject2_sampled_y1 st)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1318	(signing_attempt_state_reject2_sampled_y2 st).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1319	proof. by rewrite /signing_attempt_state_reject2_aborts. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1320	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1321	lemma signing_attempt_state_reject2_accepts_from_branch_clear md st :	States checked lemma signing_attempt_state_reject2_accepts_from_branch_clear. The following proof script must derive this proposition from prior definitions and lemmas.
1322	! signing_attempt_state_bimodal_reject2_branch st =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1323	signing_attempt_state_reject2_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1324	proof.	Begins the proof script for the preceding proposition.
1325	move=> branch_clear.	Introduces universally quantified variables or hypotheses into the proof context.
1326	by rewrite /signing_attempt_state_reject2_accepts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1327	/signing_attempt_state_reject2_aborts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1328	<code>/signing_attempt_reject2_concrete_aborts</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1329	<code>branch_clear.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1330	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1331	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1332	<code>lemma signing_attempt_state_of_sample_pair_reject2_branch_clear</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_reject2_branch_clear</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1333	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1334	<code>! signing_attempt_state_bimodal_reject2_branch</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1335	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1336	<code>proof.</code>	Begins the proof script for the preceding proposition.
1337	<code>by rewrite /signing_attempt_state_bimodal_reject2_branch</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1338	<code>/signing_attempt_state_reject2_branch_bit</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1339	<code>/signing_attempt_state_reject2_branch_byte</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1340	<code>/signature_rejection_branch_byte</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1341	<code>signing_attempt_state_of_sample_pair_signatureE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1342	<code>/sign_internal.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1343	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1344	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1345	<code>lemma signing_attempt_state_of_sample_pair_reject2_accepts</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_reject2_accepts</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1346	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1347	<code>signing_attempt_state_reject2_accepts md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1348	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1349	<code>proof.</code>	Begins the proof script for the preceding proposition.
1350	<code>apply signing_attempt_state_reject2_accepts_from_branch_clear.</code>	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1351	<code>by apply signing_attempt_state_of_sample_pair_reject2_branch_clear.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1352	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1353	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1354	<code>lemma signing_attempt_state_of_sample_pair_reject2_no_abort</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_reject2_no_abort</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1355	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1356	<code>! signing_attempt_state_reject2_aborts md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1357	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1358	<code>proof.</code>	Begins the proof script for the preceding proposition.
1359	<code>rewrite /signing_attempt_state_reject2_aborts</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1360	<code>/signing_attempt_reject2_concrete_aborts.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1361	<code>by rewrite signing_attempt_state_of_sample_pair_reject2_branch_clear.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.

Line	EasyCrypt source	Mathematical reading
1362	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1363	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1364	<code>lemma dexact_hyperball_signing_attempt_state_reject2_abort_zero</code>	States checked lemma <code>dexact_hyperball_signing_attempt_state_reject2_abort_zero</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1365	<code>md sk m ctx :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1366	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1367	<code>(signing_attempt_state_reject2_aborts md) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1368	<code>0%r.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1369	<code>proof.</code>	Begins the proof script for the preceding proposition.
1370	<code>have le0 :</code>	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1371	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1372	<code>(signing_attempt_state_reject2_aborts md) <=</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1373	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx) pred0.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1374	<code>+ apply mu_le => st_st_supp abort.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1375	<code>move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1376	<code>move=> [coins sample [_ [_ stE]]].</code>	Introduces universally quantified variables or hypotheses into the proof context.
1377	<code>move: abort.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1378	<code>rewrite stE signing_attempt_state_of_sample_pair_reject2_no_abort.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1379	<code>by [].</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1380	<code>rewrite mu0 in le0.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1381	<code>by smt(mu_bounded).</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1382	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1383	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1384	<code>lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_reduced_bound</code>	States checked lemma <code>dexact_hyperball_signing_attempt_state_reference_rejection_abort_reduced_bound</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1385	<code>md sk m ctx :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1386	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1387	<code>(signing_attempt_state_reference_rejection_aborts md) <=</code>	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1388	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1389	<code>(signing_attempt_state_reject1_aborts md) +</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1390	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1391	<code>(signing_attempt_state_hint_norm_aborts md).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1392	<code>proof.</code>	Begins the proof script for the preceding proposition.
1393	<code>have ub :=</code>	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1394	<code>dexact_hyperball_signing_attempt_state_reference_rejection_abort_union_bound</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1395	md sk m ctx.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1396	rewrite dexact_hyperball_signing_attempt_state_reject2_abort_zero	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1397	dexact_hyperball_signing_attempt_state_pack_abort_zero in ub.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1398	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1399	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1400	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1401	lemma signing_attempt_state_of_sample_pair_response_norm_accepts	States checked lemma signing_attempt_state_of_sample_pair_response_norm_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
1402	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1403	signing_attempt_state_response_norm_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1404	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1405	proof.	Begins the proof script for the preceding proposition.
1406	by rewrite /signing_attempt_state_response_norm_accepts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1407	signing_attempt_state_of_sample_pair_signatureE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1408	response_norm_ok_current.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1409	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1410	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1411	lemma signing_attempt_state_of_sample_pair_reject1_accepts	States checked lemma signing_attempt_state_of_sample_pair_reject1_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
1412	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1413	signing_attempt_state_reject1_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1414	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1415	proof.	Begins the proof script for the preceding proposition.
1416	apply signing_attempt_state_reject1_accepts_from_response_norm_ok.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1417	by apply signing_attempt_state_of_sample_pair_response_norm_accepts.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1418	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1419	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1420	lemma signing_attempt_state_of_sample_pair_reject1_no_abort	States checked lemma signing_attempt_state_of_sample_pair_reject1_no_abort. The following proof script must derive this proposition from prior definitions and lemmas.
1421	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1422	! signing_attempt_state_reject1_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1423	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1424	proof.	Begins the proof script for the preceding proposition.
1425	by rewrite /signing_attempt_state_reject1_aborts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1426	signing_attempt_state_of_sample_pair_reject1_accepts.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1427	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1428	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
1429	lemma dexact_hyperball_signing_attempt_state_reject1_abort_zero	States checked lemma dexact_hyperball_signing_attempt_state_reject1_abort_zero. The following proof script must derive this proposition from prior definitions and lemmas.
1430	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1431	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1432	(signing_attempt_state_reject1_aborts md) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1433	0%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1434	proof.	Begins the proof script for the preceding proposition.
1435	have le0 :	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1436	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1437	(signing_attempt_state_reject1_aborts md) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1438	mu (dexact_hyperball_signing_attempt_state md sk m ctx) pred0.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1439	+ apply mu_le => st st_supp abort.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1440	move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1441	move=> [coins sample [_ [_ stE]]].	Introduces universally quantified variables or hypotheses into the proof context.
1442	move: abort.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1443	rewrite stE signing_attempt_state_of_sample_pair_reject1_no_abort.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1444	by [].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1445	rewrite mu0 in le0.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1446	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1447	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1448	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1449	lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_hint_bound	States checked lemma dex- act_hyperball_signing_attempt_state_reference_rejection_abort_hint_bound. The following proof script must derive this proposition from prior definitions and lemmas.
1450	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1451	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1452	(signing_attempt_state_reference_rejection_aborts md) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1453	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1454	(signing_attempt_state_hint_norm_aborts md).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1455	proof.	Begins the proof script for the preceding proposition.
1456	have ub :=	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1457	dexact_hyperball_signing_attempt_state_reference_rejection_abort_reduced_bound	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1458	md sk m ctx.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1459	rewrite dexact_hyperball_signing_attempt_state_reject1_abort_zero in ub.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1460	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1461	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.

Line	EasyCrypt source	Mathematical reading
1462	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1463	lemma signing_attempt_state_of_sample_pair_hint_norm_accepts	States checked lemma signing_attempt_state_of_sample_pair_hint_norm_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
1464	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1465	signing_attempt_state_hint_norm_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1466	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1467	proof.	Begins the proof script for the preceding proposition.
1468	by rewrite /signing_attempt_state_hint_norm_accepts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1469	signing_attempt_state_of_sample_pair_signatureE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1470	verify_norm_ok_current.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1471	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1472	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1473	lemma signing_attempt_state_of_sample_pair_hint_norm_no_abort	States checked lemma signing_attempt_state_of_sample_pair_hint_norm_no_abort. The following proof script must derive this proposition from prior definitions and lemmas.
1474	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1475	! signing_attempt_state_hint_norm_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1476	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1477	proof.	Begins the proof script for the preceding proposition.
1478	by rewrite /signing_attempt_state_hint_norm_aborts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1479	signing_attempt_state_of_sample_pair_hint_norm_accepts.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1480	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1481	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1482	lemma dexact_hyperball_signing_attempt_state_hint_norm_abort_zero	States checked lemma dexact_hyperball_signing_attempt_state_hint_norm_abort_zero. The following proof script must derive this proposition from prior definitions and lemmas.
1483	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1484	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1485	(signing_attempt_state_hint_norm_aborts md) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1486	0%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1487	proof.	Begins the proof script for the preceding proposition.
1488	have le0 :	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1489	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1490	(signing_attempt_state_hint_norm_aborts md) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1491	mu (dexact_hyperball_signing_attempt_state md sk m ctx) pred0.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1492	+ apply mu_le => st st_supp abort.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1493	move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1494	move=> [coins sample [_ [_ stE]]].	Introduces universally quantified variables or hypotheses into the proof context.

Line	EasyCrypt source	Mathematical reading
1495	<code>move: abort.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1496	<code>rewrite stE signing_attempt_state_of_sample_pair_hint_norm_no_abort.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1497	<code>by [].</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1498	<code>rewrite mu0 in le0.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1499	<code>by smt(mu_bounded).</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1500	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1501	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1502	<code>lemma dexact_hyperball_signing_attempt_state_reference_rejection_abort_zero</code>	States checked lemma <code>dexact_hyperball_signing_attempt_state_reference_rejection_abort_zero</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1503	<code>md sk m ctx :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1504	<code>mu (dexact_hyperball_signing_attempt_state md sk m ctx)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1505	<code>(signing_attempt_state_reference_rejection_aborts md) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1506	<code>0%r.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1507	<code>proof.</code>	Begins the proof script for the preceding proposition.
1508	<code>have ub :=</code>	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
1509	<code>dexact_hyperball_signing_attempt_state_reference_rejection_abort_hint_bound</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1510	<code>md sk m ctx.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1511	<code>rewrite dexact_hyperball_signing_attempt_state_hint_norm_abort_zero in ub.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1512	<code>by smt(mu_bounded).</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1513	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1514	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1515	<code>lemma signing_attempt_state_of_coins_reject2_branch_clear_current</code>	States checked lemma <code>signing_attempt_state_of_coins_reject2_branch_clear_current</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1516	<code>md sk m ctx coins :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1517	<code>! signing_attempt_state_bimodal_reject2_branch</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1518	<code>(signing_attempt_state_of_coins md sk m ctx coins).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1519	<code>proof.</code>	Begins the proof script for the preceding proposition.
1520	<code>by rewrite /signing_attempt_state_bimodal_reject2_branch</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1521	<code>/signing_attempt_state_reject2_branch_bit</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1522	<code>/signing_attempt_state_reject2_branch_byte</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1523	<code>/signature_rejection_branch_byte</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1524	<code>signing_attempt_state_of_coins_signatureE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1525	<code>/sign_internal.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1526	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1527	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
1528	lemma signing_attempt_state_of_coins_reject2_sampled_y1E md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_reject2_sampled_y1E. The following proof script must derive this proposition from prior definitions and lemmas.
1529	signing_attempt_state_reject2_sampled_y1	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1530	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1531	signing_sample_y1 md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1532	proof.	Begins the proof script for the preceding proposition.
1533	by rewrite /signing_attempt_state_reject2_sampled_y1	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1534	signing_attempt_state_of_coins_sampled_y1E.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1535	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1536	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1537	lemma signing_attempt_state_of_coins_reject2_sampled_y2E md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_reject2_sampled_y2E. The following proof script must derive this proposition from prior definitions and lemmas.
1538	signing_attempt_state_reject2_sampled_y2	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1539	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1540	signing_sample_y2 md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1541	proof.	Begins the proof script for the preceding proposition.
1542	by rewrite /signing_attempt_state_reject2_sampled_y2	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1543	signing_attempt_state_of_coins_sampled_y2E.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1544	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1545	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1546	lemma signing_attempt_state_of_coins_reject2_balance_norm_sqE	States checked lemma signing_attempt_state_of_coins_reject2_balance_norm_sqE. The following proof script must derive this proposition from prior definitions and lemmas.
1547	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1548	signing_attempt_state_balance_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1549	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1550	signing_attempt_reject2_balance_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1551	(response_vector md sk m ctx coins)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1552	(response_aux_vector md sk m ctx coins)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1553	(signing_sample_y1 md sk m ctx coins)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1554	(signing_sample_y2 md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1555	proof.	Begins the proof script for the preceding proposition.
1556	by rewrite signing_attempt_state_balance_norm_sq_concreteE	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1557	signing_attempt_state_of_coins_responseE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1558	signing_attempt_state_of_coins_response_auxE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1559	signing_attempt_state_of_coins_reject2_sampled_y1E	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1560	signing_attempt_state_of_coins_reject2_sampled_y2E.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1561	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.

Line	EasyCrypt source	Mathematical reading
1562	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1563	lemma signing_attempt_state_of_coins_balance_norm_sq_current	States checked lemma signing_attempt_state_of_coins_balance_norm_sq_current. The following proof script must derive this proposition from prior definitions and lemmas.
1564	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1565	signing_attempt_state_balance_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1566	(signing_attempt_state_of_coins md sk m ctx coins) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1567	polyvecl_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1568	(signing_attempt_state_reject2_balance_response md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1569	(signing_attempt_state_of_coins md sk m ctx coins)) +	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1570	polyveck_norm_sq md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1571	(signing_attempt_state_reject2_balance_aux md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1572	(signing_attempt_state_of_coins md sk m ctx coins)).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1573	proof.	Begins the proof script for the preceding proposition.
1574	by rewrite /signing_attempt_state_balance_norm_sq	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1575	/signing_attempt_reject2_balance_norm_sq	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1576	/signing_attempt_state_reject2_balance_response	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1577	/signing_attempt_state_reject2_balance_aux.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1578	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1579	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1580	lemma signing_attempt_state_of_coins_reject2_accepts_current	States checked lemma signing_attempt_state_of_coins_reject2_accepts_current. The following proof script must derive this proposition from prior definitions and lemmas.
1581	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1582	signing_attempt_state_reject2_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1583	(signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1584	proof.	Begins the proof script for the preceding proposition.
1585	apply signing_attempt_state_reject2_accepts_from_branch_clear.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1586	by apply signing_attempt_state_of_coins_reject2_branch_clear_current.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1587	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1588	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1589	lemma signing_attempt_state_pack_accepts_from_valid md st :	States checked lemma signing_attempt_state_pack_accepts_from_valid. The following proof script must derive this proposition from prior definitions and lemmas.
1590	signing_attempt_state_valid_signature_accepts md st =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1591	signing_attempt_state_pack_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1592	proof.	Begins the proof script for the preceding proposition.
1593	move=> valid_ok.	Introduces universally quantified variables or hypotheses into the proof context.
1594	rewrite /signing_attempt_state_pack_accepts valid_ok.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1595	split=> //.	Splits a conjunction or structured goal into independent proof obligations.

Line	EasyCrypt source	Mathematical reading
1596	by apply mode_crypto_bytes_gt0.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1597	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1598	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1599	lemma signing_attempt_state_reference_rejection_accepts_from_accepts md st :	States checked lemma signing_attempt_state_reference_rejection_accepts_from_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
1600	signing_attempt_state_accepts md st =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1601	signing_attempt_state_reject2_accepts md st =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1602	signing_attempt_state_reference_rejection_accepts md st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1603	proof.	Begins the proof script for the preceding proposition.
1604	rewrite /signing_attempt_state_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1605	move=> [valid_ok [response_ok hint_ok]] reject2_ok.	Introduces universally quantified variables or hypotheses into the proof context.
1606	rewrite /signing_attempt_state_reference_rejection_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1607	split.	Splits a conjunction or structured goal into independent proof obligations.
1608	+ by apply signing_attempt_state_reject1_accepts_from_response_norm_ok.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1609	split.	Splits a conjunction or structured goal into independent proof obligations.
1610	+ by apply reject2_ok.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1611	split.	Splits a conjunction or structured goal into independent proof obligations.
1612	+ by apply signing_attempt_state_pack_accepts_from_valid.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1613	by apply hint_ok.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1614	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1615	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1616	lemma signing_attempt_state_of_coins_reference_rejection_accepts_current	States checked lemma signing_attempt_state_of_coins_reference_rejection_accepts_current. The following proof script must derive this proposition from prior definitions and lemmas.
1617	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1618	signing_attempt_state_reference_rejection_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1619	(signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1620	proof.	Begins the proof script for the preceding proposition.
1621	apply signing_attempt_state_reference_rejection_accepts_from_accepts.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1622	+ by apply signing_attempt_state_of_coins_accepts_current.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1623	by apply signing_attempt_state_of_coins_reject2_accepts_current.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1624	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1625	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1626	lemma signing_attempt_state_of_coins_verification_accepts_current	States checked lemma signing_attempt_state_of_coins_verification_accepts_current. The following proof script must derive this proposition from prior definitions and lemmas.
1627	md sk m ctx coins :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1628	signing_attempt_state_verification_accepts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1629	m ctx (signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1630	proof.	Begins the proof script for the preceding proposition.
1631	rewrite /signing_attempt_state_verification_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1632	split.	Splits a conjunction or structured goal into independent proof obligations.
1633	+ by apply signing_attempt_state_of_coins_field_accepts.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1634	by apply signing_attempt_state_of_coins_accepts_current.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1635	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1636	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1637	lemma signing_attempt_state_of_coins_rejection_accepts_current	States checked lemma
1638	md sk m ctx coins :	signing_attempt_state_of_coins_rejection_accepts_current. The following proof script must derive this proposition from prior definitions and lemmas.
1639	signing_attempt_state_rejection_accepts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1640	m ctx (signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1641	proof.	Begins the proof script for the preceding proposition.
1642	rewrite /signing_attempt_state_rejection_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1643	split.	Splits a conjunction or structured goal into independent proof obligations.
1644	+ by apply signing_attempt_state_of_coins_verification_accepts_current.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1645	by apply signing_attempt_state_of_coins_reference_rejection_accepts_current.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1646	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1647	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1648	lemma signing_attempt_state_of_coins_rejection_no_abort_current	States checked lemma
1649	md sk m ctx coins :	signing_attempt_state_of_coins_rejection_no_abort_current. The following proof script must derive this proposition from prior definitions and lemmas.
1650	! signing_attempt_state_rejection_aborts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1651	m ctx (signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1652	proof.	Begins the proof script for the preceding proposition.
1653	by rewrite /signing_attempt_state_rejection_aborts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1654	signing_attempt_state_of_coins_rejection_accepts_current.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1655	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1656	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1657	lemma signing_attempt_state_of_coins_no_abort_current md sk m ctx coins :	States checked lemma signing_attempt_state_of_coins_no_abort_current. The following proof script must derive this proposition from prior definitions and lemmas.
1658	! signing_attempt_state_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1659	(signing_attempt_state_of_coins md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1660	proof.	Begins the proof script for the preceding proposition.
1661	by rewrite /signing_attempt_state_aborts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1662	/signing_attempt_state_rejection_sampling_aborts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1663	/signing_attempt_state_response_norm_aborts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1664	/signing_attempt_state_hint_norm_aborts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1665	/signing_attempt_state_valid_signature_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1666	/signing_attempt_state_response_norm_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1667	/signing_attempt_state_hint_norm_accepts	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1668	signing_attempt_state_of_coins_signatureE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1669	valid_signature_sign_internal	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1670	response_norm_ok_current	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1671	verify_norm_ok_current.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1672	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1673	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1674	lemma signing_attempt_state_distribution_lossless md sk m ctx :	States checked lemma <code>signing_attempt_state_distribution_lossless</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1675	is_lossless (dsigning_attempt_state md sk m ctx).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1676	proof.	Begins the proof script for the preceding proposition.
1677	rewrite /dsigning_attempt_state.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1678	by apply dmap_ll; apply signing_coin_distribution_lossless.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1679	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1680	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1681	lemma signing_attempt_state_distribution_token_spread	States checked lemma <code>signing_attempt_state_distribution_token_spread</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1682	md sk m ctx token :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1683	mu (dsigning_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1684	(fun st =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1685	signing_entropy_token_of_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1686	(signing_attempt_state_coins st) = token) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1687	signing_randomness_point_bound.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1688	proof.	Begins the proof script for the preceding proposition.
1689	rewrite /dsigning_attempt_state dmapE /=.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1690	apply (1er_trans	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1691	(mu drandom_coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1692	(fun coins => signing_entropy_token_of_coins coins = token))).	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1693	+ apply mu_le => coins _.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1694	by rewrite /signing_attempt_state_coins /signing_attempt_state_of_coins.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1695	by apply signing_coin_distribution_token_point_bound.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1696	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1697	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1698	lemma signing_attempt_state_distribution_reference_rejection_abort_zero_current	States checked lemma <code>signing_attempt_state_distribution_reference_rejection_abort_zero_current</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1699	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1700	mu (dsigning_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1701	(signing_attempt_state_reference_rejection_aborts md) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1702	0%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1703	proof.	Begins the proof script for the preceding proposition.
1704	rewrite /dsigning_attempt_state dmapE /=.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1705	rewrite (mu_eq _ _ (fun (_ : random_coins) => false)).	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1706	+ move=> coins.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1707	rewrite /signing_attempt_state_reference_rejection_aborts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1708	by smt(signing_attempt_state_of_coins_reference_rejection_accepts_current).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1709	by rewrite mu0.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1710	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1711	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1712	lemma signing_attempt_state_distribution_rejection_abort_zero_current	States checked lemma signing_attempt_state_distribution_rejection_abort_zero_current. The following proof script must derive this proposition from prior definitions and lemmas.
1713	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1714	mu (dsigning_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1715	(signing_attempt_state_rejection_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1716	(public_key_of_secret md sk) m ctx) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1717	0%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1718	proof.	Begins the proof script for the preceding proposition.
1719	rewrite /dsigning_attempt_state dmapE /=.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1720	rewrite (mu_eq _ _ (fun (_ : random_coins) => false)).	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1721	+ move=> coins.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1722	by smt(signing_attempt_state_of_coins_rejection_no_abort_current).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1723	by rewrite mu0.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1724	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1725	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1726	lemma sign_internal_simulation_relation md sk m ctx coins :	States checked lemma sign_internal_simulation_relation. The following proof script must derive this proposition from prior definitions and lemmas.
1727	signing_simulation_relation md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1728	(public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1729	(sign_internal md sk m ctx coins).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1730	proof.	Begins the proof script for the preceding proposition.
1731	by apply (signing_attempt_simulates md (public_key_of_secret md sk)	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1732	sk m ctx coins (sign_internal md sk m ctx coins));	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1733	apply sign_internal_attempt_relation.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1734	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.

Line	EasyCrypt source	Mathematical reading
1735	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1736	lemma signing_attempt_transcript_valid md pk sk m ctx coins sig :	States checked lemma <code>signing_attempt_transcript_valid</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1737	signing_attempt_relation md pk sk m ctx coins sig =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1738	transcript_valid md (transcript_of_signature md pk m ctx sig).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1739	proof.	Begins the proof script for the preceding proposition.
1740	rewrite /signing_attempt_relation /valid_keypair.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1741	move=> [#] pkE _ ->.	Introduces universally quantified variables or hypotheses into the proof context.
1742	rewrite /transcript_valid /transcript_verifies	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1743	/transcript_of_signature /transcript_pk /transcript_message	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1744	/transcript_context /transcript_signature /=.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1745	split.	Splits a conjunction or structured goal into independent proof obligations.
1746	+ rewrite pkE.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1747	by apply verify_internal_sign_internal.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1748	split.	Splits a conjunction or structured goal into independent proof obligations.
1749	+ by apply valid_signature_sign_internal.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1750	by rewrite /transcript_fields_match_signature /transcript_of_signature /=.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1751	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1752	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1753	lemma signing_attempt_transcript_relation md pk sk m ctx coins sig :	States checked lemma <code>signing_attempt_transcript_relation</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1754	signing_attempt_relation md pk sk m ctx coins sig =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1755	signing_transcript_relation md pk m ctx sig	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1756	(transcript_of_signature md pk m ctx sig).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1757	proof.	Begins the proof script for the preceding proposition.
1758	move=> attempt.	Introduces universally quantified variables or hypotheses into the proof context.
1759	rewrite /signing_transcript_relation.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1760	split.	Splits a conjunction or structured goal into independent proof obligations.
1761	+ by apply (signing_attempt_simulates md pk sk m ctx coins sig).	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1762	split=> //.	Splits a conjunction or structured goal into independent proof obligations.
1763	by apply (signing_attempt_transcript_valid md pk sk m ctx coins sig).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1764	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1765	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1766	lemma sign_internal_transcript_relation md sk m ctx coins :	States checked lemma <code>sign_internal_transcript_relation</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1767	signing_transcript_relation md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1768	(public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1769	(sign_internal md sk m ctx coins)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1770	(transcript_of_signature md (public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1771	(sign_internal md sk m ctx coins)).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1772	proof.	Begins the proof script for the preceding proposition.
1773	by apply (signing_attempt_transcript_relation md	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1774	(public_key_of_secret md sk) sk m ctx coins	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1775	(sign_internal md sk m ctx coins));	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1776	apply sign_internal_attempt_relation.	Applies an existing theorem, lemma, or proof rule to reduce the current goal to its premises.
1777	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1778	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1779	lemma sign_internal_record_relation md sk m ctx coins :	States checked lemma <code>sign_internal_record_relation</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1780	signing_record_relation md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1781	(m, ctx,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1782	sign_internal md sk m ctx coins,	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1783	transcript_of_signature md (public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1784	(sign_internal md sk m ctx coins)).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1785	proof.	Begins the proof script for the preceding proposition.
1786	rewrite /signing_record_relation /signing_record_message	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1787	/signing_record_context /signing_record_signature	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1788	/signing_record_transcript /=.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1789	by apply sign_internal_transcript_relation.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1790	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1791	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1792	lemma signing_record_relation_sound md pk r :	States checked lemma <code>signing_record_relation_sound</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1793	signing_record_relation md pk r =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1794	signing_simulation_relation md pk	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1795	(signing_record_message r)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1796	(signing_record_context r)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1797	(signing_record_signature r) /\	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1798	transcript_valid md (signing_record_transcript r).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1799	proof.	Begins the proof script for the preceding proposition.
1800	rewrite /signing_record_relation /signing_transcript_relation.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1801	move=> h.	Introduces universally quantified variables or hypotheses into the proof context.
1802	case: h => hsim rest.	Performs case analysis or induction, splitting the mathematical proof by constructors or list structure.
1803	case: rest => _ valid.	Performs case analysis or induction, splitting the mathematical proof by constructors or list structure.
1804	split.	Splits a conjunction or structured goal into independent proof obligations.
1805	+ by apply hsim.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1806	by apply valid.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1807	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1808	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
1809	lemma signing_record_log_sound_cons md pk r rs :	States checked lemma <code>signing_record_log_sound_cons</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1810	signing_record_relation md pk r =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1811	signing_record_log_sound md pk rs =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1812	signing_record_log_sound md pk (r :: rs).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1813	proof.	Begins the proof script for the preceding proposition.
1814	by rewrite /signing_record_log_sound /.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1815	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1816	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1817	lemma signing_record_log_sound_transcripts_valid md pk rs :	States checked lemma <code>signing_record_log_sound_transcripts_valid</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1818	signing_record_log_sound md pk rs =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1819	transcript_log_valid md (signing_record_transcripts rs).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1820	proof.	Begins the proof script for the preceding proposition.
1821	rewrite /signing_record_log_sound /transcript_log_valid	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1822	/signing_record_transcripts.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1823	elim: rs => [!r rs ih] //.	Performs case analysis or induction, splitting the mathematical proof by constructors or list structure.
1824	move=> [rel sound].	Introduces universally quantified variables or hypotheses into the proof context.
1825	split.	Splits a conjunction or structured goal into independent proof obligations.
1826	+ by case: (signing_record_relation_sound md pk r rel).	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1827	by apply ih.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1828	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1829	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1830	lemma signing_record_queries_cons r rs :	States checked lemma <code>signing_record_queries_cons</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1831	signing_record_queries (r :: rs) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1832	signing_record_query r :: signing_record_queries rs.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1833	proof. by rewrite /signing_record_queries. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1834	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1835	lemma signing_record_transcripts_cons r rs :	States checked lemma <code>signing_record_transcripts_cons</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1836	signing_record_transcripts (r :: rs) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1837	signing_record_transcript r :: signing_record_transcripts rs.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1838	proof. by rewrite /signing_record_transcripts. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1839	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1840	lemma rejection_bound_parts_nonnegative :	States checked lemma <code>rejection_bound_parts_nonnegative</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1841	rejection_sampling_bound_obligation =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
1842	0%r <= rejection_sampling_loss_term.	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
1843	proof. by rewrite /rejection_sampling_bound_obligation. qed.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1844	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1845	lemma rejection_sampling_bound_obligation_holds :	States checked lemma rejection_sampling_bound_obligation_holds. The following proof script must derive this proposition from prior definitions and lemmas.
1846	rejection_sampling_bound_obligation.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1847	proof.	Begins the proof script for the preceding proposition.
1848	rewrite /rejection_sampling_bound_obligation.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
1849	split.	Splits a conjunction or structured goal into independent proof obligations.
1850	+ by apply rejection_sampling_loss_term_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1851	split.	Splits a conjunction or structured goal into independent proof obligations.
1852	+ by apply fs_with_aborts_reprogramming_term_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1853	+ by apply fs_with_aborts_min_entropy_term_nonnegative.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
1854	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1855	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1856	lemma signing_attempt_state_of_sample_pair_sampled_yE md sk m ctx coins sample :	States checked lemma signing_attempt_state_of_sample_pair_sampled_yE. The following proof script must derive this proposition from prior definitions and lemmas.
1857	signing_attempt_state_sampled_y	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1858	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1859	commitment_raw md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1860	proof.	Begins the proof script for the preceding proposition.
1861	by rewrite /signing_attempt_state_sampled_y	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1862	/signing_attempt_sample_commitment_raw	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1863	/signing_attempt_state_sample	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1864	/signing_attempt_state_of_sample_pair	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1865	/signing_attempt_sample_of_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1866	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1867	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1868	lemma signing_attempt_state_of_sample_pair_highbitsE md sk m ctx coins sample :	States checked lemma signing_attempt_state_of_sample_pair_highbitsE. The following proof script must derive this proposition from prior definitions and lemmas.
1869	signing_attempt_state_highbits	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1870	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1871	commitment_highbits md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1872	proof.	Begins the proof script for the preceding proposition.
1873	by rewrite /signing_attempt_state_highbits	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1874	/signing_attempt_state_of_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1875	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1876	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1877	lemma signing_attempt_state_of_sample_pair_lowbitsE md sk m ctx coins sample :	States checked lemma signing_attempt_state_of_sample_pair_lowbitsE. The following proof script must derive this proposition from prior definitions and lemmas.
1878	signing_attempt_state_lowbits	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1879	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1880	commitment_lowbits md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1881	proof.	Begins the proof script for the preceding proposition.
1882	by rewrite /signing_attempt_state_lowbits	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1883	/signing_attempt_state_of_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1884	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1885	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1886	lemma signing_attempt_state_of_sample_pair_challengeE md sk m ctx coins sample :	States checked lemma signing_attempt_state_of_sample_pair_challengeE. The following proof script must derive this proposition from prior definitions and lemmas.
1887	signing_attempt_state_challenge	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1888	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1889	commitment_challenge md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1890	proof.	Begins the proof script for the preceding proposition.
1891	by rewrite /signing_attempt_state_challenge	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1892	/signing_attempt_state_of_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1893	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1894	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1895	lemma signing_attempt_state_of_sample_pair_responseE md sk m ctx coins sample :	States checked lemma signing_attempt_state_of_sample_pair_responseE. The following proof script must derive this proposition from prior definitions and lemmas.
1896	signing_attempt_state_response	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1897	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1898	response_vector md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1899	proof.	Begins the proof script for the preceding proposition.
1900	by rewrite /signing_attempt_state_response	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1901	/signing_attempt_state_of_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1902	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1903	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1904	lemma signing_attempt_state_of_sample_pair_response_auxE	States checked lemma signing_attempt_state_of_sample_pair_response_auxE. The following proof script must derive this proposition from prior definitions and lemmas.
1905	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1906	signing_attempt_state_response_aux	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1907	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1908	response_aux_vector md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1909	proof.	Begins the proof script for the preceding proposition.
1910	by rewrite /signing_attempt_state_response_aux	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1911	/signing_attempt_state_of_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1912	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1913	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
1914	lemma signing_attempt_state_of_sample_pair_hintE md sk m ctx coins sample :	States checked lemma <code>signing_attempt_state_of_sample_pair_hintE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1915	signing_attempt_state_hint	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1916	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1917	hint_vector md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1918	proof.	Begins the proof script for the preceding proposition.
1919	by rewrite /signing_attempt_state_hint	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1920	/signing_attempt_state_of_sample_pair.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1921	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1922	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1923	lemma signing_attempt_state_of_sample_pair_lowbits_carrierE	States checked lemma <code>signing_attempt_state_of_sample_pair_lowbits_carrierE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1924	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1925	signing_attempt_state_lowbits_carrier	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1926	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1927	nth poly_zero (commitment_raw md sk m ctx coins) 0.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1928	proof.	Begins the proof script for the preceding proposition.
1929	by rewrite /signing_attempt_state_lowbits_carrier	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1930	signing_attempt_state_of_sample_pair_sampled_yE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1931	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1932	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1933	lemma signing_attempt_state_of_sample_pair_tokenE md sk m ctx coins sample :	States checked lemma <code>signing_attempt_state_of_sample_pair_tokenE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1934	signing_attempt_state_token	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1935	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1936	signature_token md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1937	proof.	Begins the proof script for the preceding proposition.
1938	by rewrite /signing_attempt_state_token	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1939	signing_attempt_state_of_sample_pair_responseE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1940	signing_attempt_state_of_sample_pair_response_auxE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1941	signing_attempt_state_of_sample_pair_hintE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1942	/signature_token.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1943	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1944	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1945	lemma signing_attempt_state_of_sample_pair_programmed_lowbitsE	States checked lemma <code>signing_attempt_state_of_sample_pair_programmed_lowbitsE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1946	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1947	signing_attempt_state_programmed_lowbits	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1948	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1949	commitment_lowbits md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1950	proof.	Begins the proof script for the preceding proposition.
1951	by rewrite /signing_attempt_state_programmed_lowbits	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1952	signing_attempt_state_of_sample_pair_lowbits_carrierE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1953	/signing_attempt_state_coins /signing_attempt_state_of_sample_pair	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1954	/commitment_lowbits.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1955	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1956	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1957	lemma signing_attempt_state_of_sample_pair_lowbits_consistent	States checked lemma
1958	md sk m ctx coins sample :	signing_attempt_state_of_sample_pair_lowbits_consistent. The following proof script must derive this proposition from prior definitions and lemmas.
1959	signing_attempt_state_lowbits_consistent	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1960	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1961	proof.	Begins the proof script for the preceding proposition.
1962	by rewrite /signing_attempt_state_lowbits_consistent	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1963	signing_attempt_state_of_sample_pair_lowbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1964	signing_attempt_state_of_sample_pair_programmed_lowbitsE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1965	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1966	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1967	lemma signing_attempt_state_of_sample_pair_programmed_challengeE	States checked lemma
1968	md sk m ctx coins sample :	signing_attempt_state_of_sample_pair_programmed_challengeE. The following proof script must derive this proposition from prior definitions and lemmas.
1969	signing_attempt_state_programmed_challenge md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1970	m ctx (signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1971	commitment_challenge md sk m ctx coins.	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1972	proof.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1973	by rewrite /signing_attempt_state_programmed_challenge	Begins the proof script for the preceding proposition.
1974	signing_attempt_state_of_sample_pair_response_auxE	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1975	signing_attempt_state_of_sample_pair_lowbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1976	/commitment_challenge.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1977	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
1978	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1979	lemma signing_attempt_state_of_sample_pair_challenge_consistent	States checked lemma
1980	md sk m ctx coins sample :	signing_attempt_state_of_sample_pair_challenge_consistent. The following proof script must derive this proposition from prior definitions and lemmas.
1981	signing_attempt_state_challenge_consistent md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
1982	<code>m ctx (signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1983	<code>proof.</code>	Begins the proof script for the preceding proposition.
1984	<code>by rewrite /signing_attempt_state_challenge_consistent</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1985	<code>signing_attempt_state_of_sample_pair_challengeE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1986	<code>signing_attempt_state_of_sample_pair_programmed_challengeE.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1987	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
1988	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
1989	<code>lemma signing_attempt_state_of_sample_pair_reconstructed_highbitsE</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_reconstructed_highbitsE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
1990	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1991	<code>signing_attempt_state_reconstructed_highbits md (public_key_of_secret md sk)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1992	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =</code>	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
1993	<code>commitment_highbits md sk m ctx coins.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1994	<code>proof.</code>	Begins the proof script for the preceding proposition.
1995	<code>by rewrite /signing_attempt_state_reconstructed_highbits</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
1996	<code>signing_attempt_state_of_sample_pair_response_auxE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1997	<code>signing_attempt_state_of_sample_pair_challengeE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1998	<code>/commitment_highbits.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
1999	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2000	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2001	<code>lemma signing_attempt_state_of_sample_pair_public_equation_consistent</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_public_equation_consistent</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2002	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2003	<code>signing_attempt_state_public_equation_consistent md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2004	<code>(public_key_of_secret md sk)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2005	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2006	<code>proof.</code>	Begins the proof script for the preceding proposition.
2007	<code>by rewrite /signing_attempt_state_public_equation_consistent</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2008	<code>signing_attempt_state_of_sample_pair_highbitsE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2009	<code>signing_attempt_state_of_sample_pair_reconstructed_highbitsE.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2010	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2011	<code>blank</code>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2012	<code>lemma signing_attempt_state_of_sample_pair_signature_of_fieldsE</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_signature_of_fieldsE</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2013	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2014	<code>signing_attempt_state_signature_of_fields</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
2015	(public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2016	(signing_attempt_state_of_sample_pair md sk m ctx coins sample) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.
2017	sign_internal md sk m ctx coins.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2018	proof.	Begins the proof script for the preceding proposition.
2019	by rewrite /signing_attempt_state_signature_of_fields /sign_internal	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2020	signing_attempt_state_of_sample_pair_highbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2021	signing_attempt_state_of_sample_pair_lowbitsE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2022	signing_attempt_state_of_sample_pair_challengeE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2023	signing_attempt_state_of_sample_pair_tokenE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2024	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2025	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2026	lemma signing_attempt_state_of_sample_pair_fields_match_signature	States checked lemma signing_attempt_state_of_sample_pair_fields_match_signature. The following proof script must derive this proposition from prior definitions and lemmas.
2027	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2028	signing_attempt_state_fields_match_signature	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2029	(public_key_of_secret md sk) m ctx	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2030	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2031	proof.	Begins the proof script for the preceding proposition.
2032	by rewrite /signing_attempt_state_fields_match_signature	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2033	signing_attempt_state_of_sample_pair_signatureE	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2034	signing_attempt_state_of_sample_pair_signature_of_fieldsE.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2035	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2036	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2037	lemma signing_attempt_state_of_sample_pair_field_accepts	States checked lemma signing_attempt_state_of_sample_pair_field_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
2038	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2039	signing_attempt_state_field_accepts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2040	m ctx (signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2041	proof.	Begins the proof script for the preceding proposition.
2042	rewrite /signing_attempt_state_field_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2043	split.	Splits a conjunction or structured goal into independent proof obligations.
2044	+ by apply signing_attempt_state_of_sample_pair_lowbits_consistent.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2045	split.	Splits a conjunction or structured goal into independent proof obligations.
2046	+ by apply signing_attempt_state_of_sample_pair_challenge_consistent.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2047	split.	Splits a conjunction or structured goal into independent proof obligations.
2048	+ by apply signing_attempt_state_of_sample_pair_public_equation_consistent.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2049	by apply signing_attempt_state_of_sample_pair_fields_match_signature.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2050	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.

Line	EasyCrypt source	Mathematical reading
2051	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2052	<code>lemma signing_attempt_state_of_sample_pair_valid_signature_accepts</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_valid_signature_accepts</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2053	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2054	<code>signing_attempt_state_valid_signature_accepts md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2055	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2056	<code>proof.</code>	Begins the proof script for the preceding proposition.
2057	<code>by rewrite /signing_attempt_state_valid_signature_accepts</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2058	<code>signing_attempt_state_of_sample_pair_signatureE</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2059	<code>valid_signature_sign_internal.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2060	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2061	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2062	<code>lemma signing_attempt_state_of_sample_pair_accepts_current</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_accepts_current</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2063	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2064	<code>signing_attempt_state_accepts md</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2065	<code>(signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2066	<code>proof.</code>	Begins the proof script for the preceding proposition.
2067	<code>rewrite /signing_attempt_state_accepts.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2068	<code>split.</code>	Splits a conjunction or structured goal into independent proof obligations.
2069	<code>+ by apply signing_attempt_state_of_sample_pair_valid_signature_accepts.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2070	<code>split.</code>	Splits a conjunction or structured goal into independent proof obligations.
2071	<code>+ by apply signing_attempt_state_of_sample_pair_response_norm_accepts.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2072	<code>by apply signing_attempt_state_of_sample_pair_hint_norm_accepts.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2073	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2074	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2075	<code>lemma signing_attempt_state_of_sample_pair_verification_accepts_current</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_verification_accepts_current</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2076	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2077	<code>signing_attempt_state_verification_accepts md (public_key_of_secret md sk)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2078	<code>m ctx (signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2079	<code>proof.</code>	Begins the proof script for the preceding proposition.
2080	<code>rewrite /signing_attempt_state_verification_accepts.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2081	<code>split.</code>	Splits a conjunction or structured goal into independent proof obligations.
2082	<code>+ by apply signing_attempt_state_of_sample_pair_field_accepts.</code>	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2083	<code>by apply signing_attempt_state_of_sample_pair_accepts_current.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2084	<code>qed.</code>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2085	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.

Line	EasyCrypt source	Mathematical reading
2086	lemma signing_attempt_state_of_sample_pair_reference_rejection_accepts	States checked lemma signing_attempt_state_of_sample_pair_reference_rejection_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
2087	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2088	(coins : random_coins) (sample : signing_sample_pair) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2089	signing_attempt_state_reference_rejection_accepts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2090	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2091	proof.	Begins the proof script for the preceding proposition.
2092	rewrite /signing_attempt_state_reference_rejection_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2093	split.	Splits a conjunction or structured goal into independent proof obligations.
2094	+ by apply signing_attempt_state_of_sample_pair_reject1_accepts.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2095	split.	Splits a conjunction or structured goal into independent proof obligations.
2096	+ by apply signing_attempt_state_of_sample_pair_reject2_accepts.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2097	split.	Splits a conjunction or structured goal into independent proof obligations.
2098	+ by apply signing_attempt_state_of_sample_pair_pack_accepts.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2099	by apply signing_attempt_state_of_sample_pair_hint_norm_accepts.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2100	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2101	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2102	lemma signing_attempt_state_of_sample_pair_reference_rejection_no_abort	States checked lemma signing_attempt_state_of_sample_pair_reference_rejection_no_abort. The following proof script must derive this proposition from prior definitions and lemmas.
2103	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2104	(coins : random_coins) (sample : signing_sample_pair) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2105	! signing_attempt_state_reference_rejection_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2106	(signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2107	proof.	Begins the proof script for the preceding proposition.
2108	by rewrite /signing_attempt_state_reference_rejection_aborts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2109	signing_attempt_state_of_sample_pair_reference_rejection_accepts.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2110	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2111	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2112	lemma signing_attempt_state_of_sample_pair_rejection_accepts_current	States checked lemma signing_attempt_state_of_sample_pair_rejection_accepts_current. The following proof script must derive this proposition from prior definitions and lemmas.
2113	md sk m ctx coins sample :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2114	signing_attempt_state_rejection_accepts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2115	m ctx (signing_attempt_state_of_sample_pair md sk m ctx coins sample).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2116	proof.	Begins the proof script for the preceding proposition.
2117	rewrite /signing_attempt_state_rejection_accepts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2118	split.	Splits a conjunction or structured goal into independent proof obligations.
2119	+ by apply signing_attempt_state_of_sample_pair_verification_accepts_current.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2120	by apply signing_attempt_state_of_sample_pair_reference_rejection_accepts.	Discharges the current goal in one step using the cited simplification, lemma, or automation.

Line	EasyCrypt source	Mathematical reading
2121	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2122	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2123	<code>lemma signing_attempt_state_of_sample_pair_rejection_no_abort_current</code>	States checked lemma <code>signing_attempt_state_of_sample_pair_rejection_no_abort_current</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2124	<code>md sk m ctx coins sample :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2125	<code>! signing_attempt_state_rejection_aborts md (public_key_of_secret md sk)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2126	<code>m ctx (signing_attempt_state_of_sample_pair md sk m ctx coins sample).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2127	<code>proof.</code>	Begins the proof script for the preceding proposition.
2128	<code>by rewrite /signing_attempt_state_rejection_aborts</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2129	<code>signing_attempt_state_of_sample_pair_rejection_accepts_current.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2130	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2131	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2132	<code>lemma dexact_hyperball_signing_attempt_state_reference_rejection_accepts</code>	States checked lemma <code>dexact_hyperball_signing_attempt_state_reference_rejection_accepts</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2133	<code>(md : mode) (sk : skey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2134	<code>(st : signing_attempt_state) :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2135	<code>st \in dexact_hyperball_signing_attempt_state md sk m ctx =></code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
2136	<code>signing_attempt_state_reference_rejection_accepts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2137	<code>proof.</code>	Begins the proof script for the preceding proposition.
2138	<code>move=> st_supp.</code>	Introduces universally quantified variables or hypotheses into the proof context.
2139	<code>move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2140	<code>move=> [coins sample [_ [stE]]].</code>	Introduces universally quantified variables or hypotheses into the proof context.
2141	<code>by rewrite stE signing_attempt_state_of_sample_pair_reference_rejection_accepts.</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2142	<i>qed.</i>	Ends the proof; EasyCrypt has accepted all remaining obligations.
2143	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2144	<code>lemma dexact_hyperball_signing_attempt_state_reference_rejection_no_abort</code>	States checked lemma <code>dexact_hyperball_signing_attempt_state_reference_rejection_no_abort</code> . The following proof script must derive this proposition from prior definitions and lemmas.
2145	<code>(md : mode) (sk : skey) (m : message) (ctx : context)</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2146	<code>(st : signing_attempt_state) :</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2147	<code>st \in dexact_hyperball_signing_attempt_state md sk m ctx =></code>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
2148	<code>! signing_attempt_state_reference_rejection_aborts md st.</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2149	<code>proof.</code>	Begins the proof script for the preceding proposition.
2150	<code>move=> st_supp.</code>	Introduces universally quantified variables or hypotheses into the proof context.
2151	<code>rewrite /signing_attempt_state_reference_rejection_aborts.</code>	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2152	<code>by rewrite (dexact_hyperball_signing_attempt_state_reference_rejection_accepts</code>	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2153	<code>md sk m ctx st st_supp).</code>	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.

Line	EasyCrypt source	Mathematical reading
2154	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2155	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2156	lemma dexact_hyperball_signing_attempt_state_rejection_accepts	States checked lemma dexact_hyperball_signing_attempt_state_rejection_accepts. The following proof script must derive this proposition from prior definitions and lemmas.
2157	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2158	(st : signing_attempt_state) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2159	st \in dexact_hyperball_signing_attempt_state md sk m ctx =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
2160	signing_attempt_state_rejection_accepts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2161	m ctx st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2162	proof.	Begins the proof script for the preceding proposition.
2163	move=> st_supp.	Introduces universally quantified variables or hypotheses into the proof context.
2164	move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2165	move=> [coins sample [_ stE]]].	Introduces universally quantified variables or hypotheses into the proof context.
2166	by rewrite stE signing_attempt_state_of_sample_pair_rejection_accepts_current.	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2167	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2168	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2169	lemma dexact_hyperball_signing_attempt_state_rejection_no_abort	States checked lemma dexact_hyperball_signing_attempt_state_rejection_no_abort. The following proof script must derive this proposition from prior definitions and lemmas.
2170	(md : mode) (sk : skey) (m : message) (ctx : context)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2171	(st : signing_attempt_state) :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2172	st \in dexact_hyperball_signing_attempt_state md sk m ctx =>	Introduces implication structure: later proof steps may assume the antecedent to prove the conclusion.
2173	! signing_attempt_state_rejection_aborts md (public_key_of_secret md sk)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2174	m ctx st.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2175	proof.	Begins the proof script for the preceding proposition.
2176	move=> st_supp.	Introduces universally quantified variables or hypotheses into the proof context.
2177	rewrite /signing_attempt_state_rejection_aborts.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2178	by rewrite (dexact_hyperball_signing_attempt_state_rejection_accepts	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2179	md sk m ctx st st_supp).	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2180	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2181	<i>blank</i>	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2182	lemma dexact_hyperball_signing_attempt_state_rejection_abort_zero	States checked lemma dexact_hyperball_signing_attempt_state_rejection_abort_zero. The following proof script must derive this proposition from prior definitions and lemmas.
2183	md sk m ctx :	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2184	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2185	(signing_attempt_state_rejection_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2186	(public_key_of_secret md sk) m ctx) =	Part of an equality or definition, identifying two mathematical expressions or assigning a program value.

Line	EasyCrypt source	Mathematical reading
2187	0%r.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2188	proof.	Begins the proof script for the preceding proposition.
2189	have le0 :	Creates an intermediate claim. This mirrors a mathematical proof that first proves a sub-inequality or invariant.
2190	mu (dexact_hyperball_signing_attempt_state md sk m ctx)	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2191	(signing_attempt_state_rejection_aborts md	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2192	(public_key_of_secret md sk) m ctx) <=	Part of an inequality, usually comparing probabilities or bounding a concrete loss term.
2193	mu (dexact_hyperball_signing_attempt_state md sk m ctx) pred0.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2194	+ apply mu_le => st st_supp abort.	Solves a proof branch produced by a previous split, transitivity, or case analysis.
2195	move: st_supp; rewrite dexact_hyperball_signing_attempt_state_supportP.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2196	move=> [coins sample [_ [_ stE]]].	Introduces universally quantified variables or hypotheses into the proof context.
2197	move: abort.	Continuation line in the current declaration or proof. Read it with surrounding lines as part of the same mathematical object.
2198	rewrite stE signing_attempt_state_of_sample_pair_rejection_no_abort_current.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2199	by [].	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2200	rewrite mu0 in le0.	Rewrites the goal using definitions or previously proved equalities, exposing the intended mathematical normal form.
2201	by smt(mu_bounded).	Discharges the current goal in one step using the cited simplification, lemma, or automation.
2202	qed.	Ends the proof; EasyCrypt has accepted all remaining obligations.
2203	blank	Blank separator. It marks a boundary between declarations, proof blocks, or tactic phases.
2204	end HAETAE_Rejection.	Closes the current theory or module block.